

FROM  EXCEL TO 



WHO ARE WE?

Center for Health Data Science (**HeaDS**):

- Data Science Research Groups
- HDS Sandbox (Infrastructure Project)
- **SUND Data Lab**

<https://heads.ku.dk/>

SUND Data Lab:

- Courses & Workshops, Seminars, etc.
- Health Data Science Consultations
- Commissioned Research & Supervision
- Hub for Data Science (Matchmaking)

Data Science Laboratory (DSL) - <https://datalab.science.ku.dk/>



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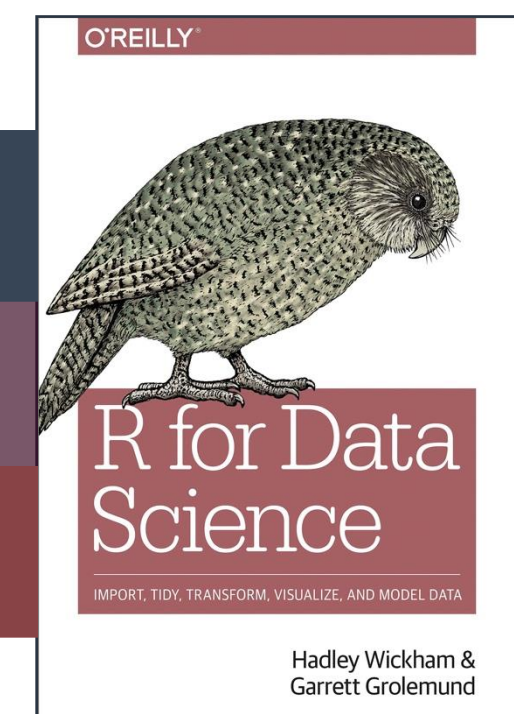
FROM EXCEL TO R

THE PRACTICALS

Two days: 9.00-16.00. There will be coffee breaks, we promise ☕

“R for Data Science” - a generally useful book on R, also for this course

The course is build on hands-on presentations & exercises

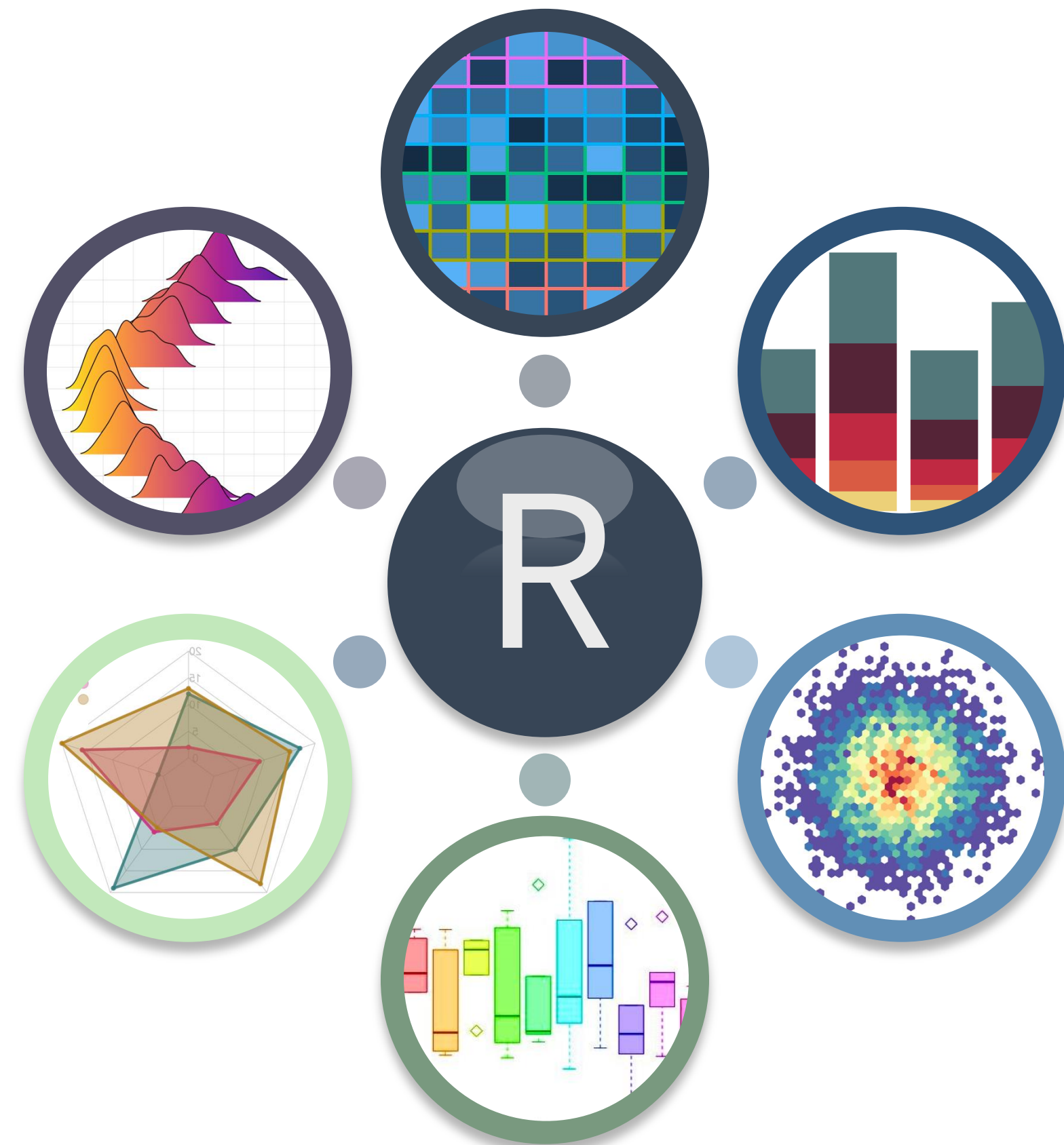
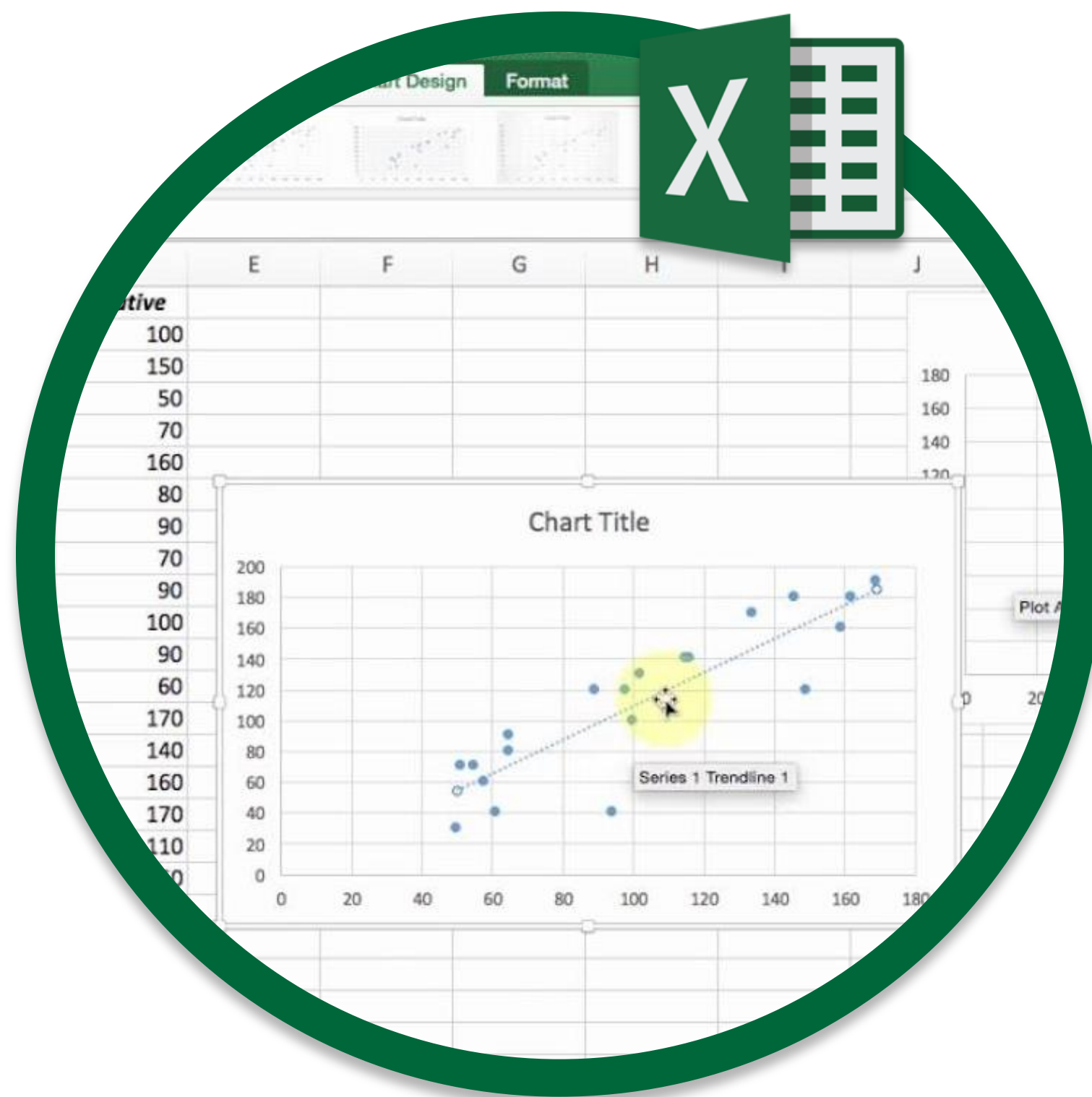


Download and install the newest version of R (<https://cran.r-project.org/>)

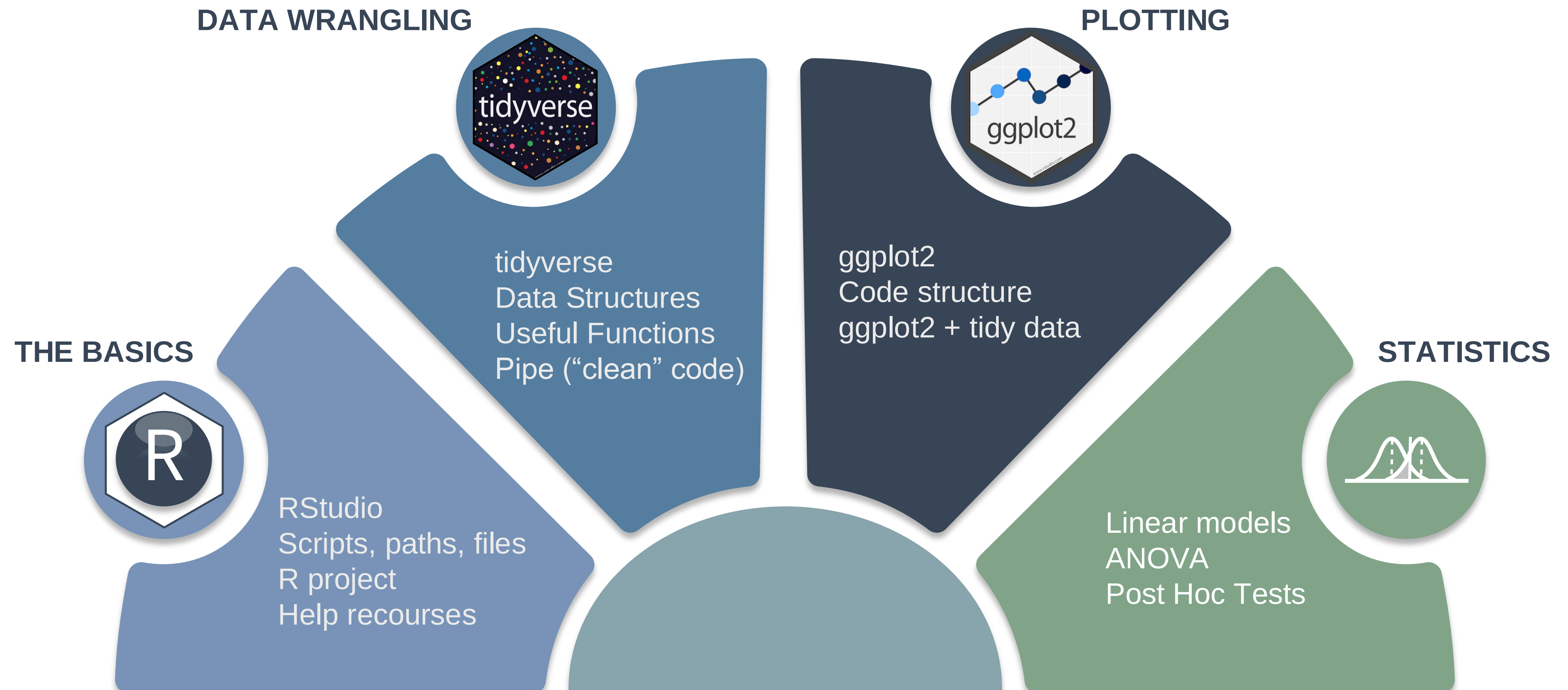
Download and install the newest version of R-studio (<https://posit.co/download/rstudio-desktop/>)

Go to course website: <https://center-for-health-data-science.github.io/FromExceltoR/>

— WELCOME TO FROM EXCEL TO R



WHAT WILL I LEARN?



PROGRAM

DAY 1

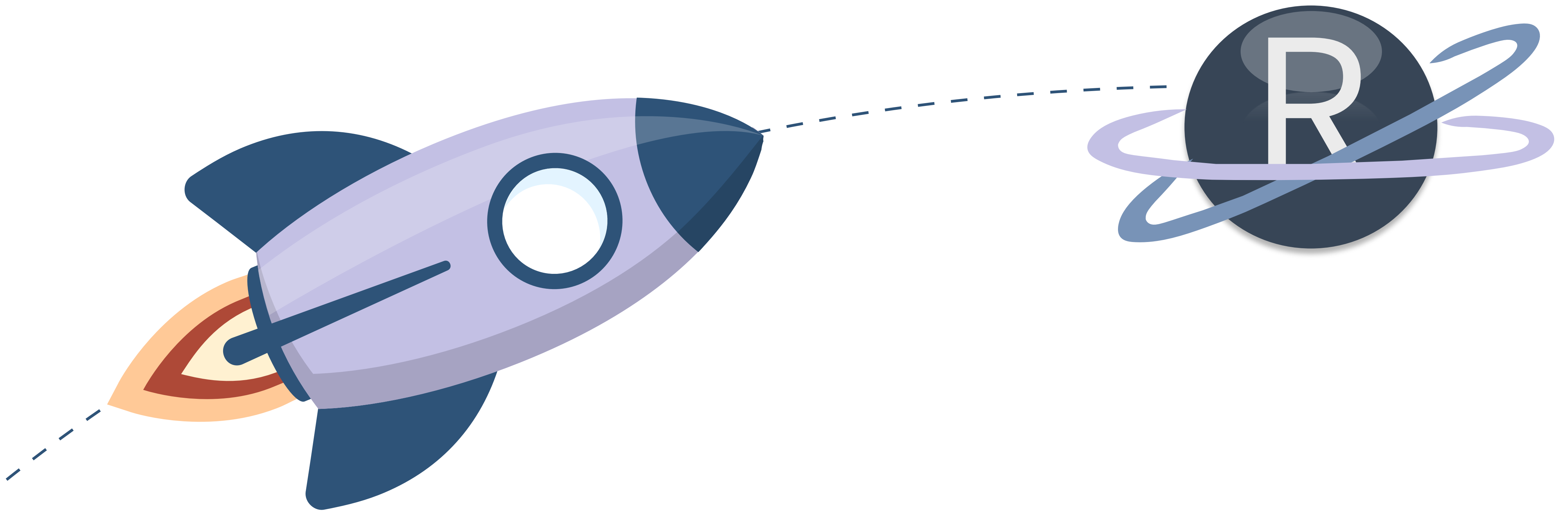
08:30 - Installation Issues + Coffee
09:00 - Introduction to R
09:45 - Exercise 1A
10:15 - Break
10:30 - Introduction to Quarto
11:00 - Exercise 1B
12:00 - Lunch
13:00 – Tidyverse
14:00 - Break
14:15 - Tidyverse Exercise
16:00 - See you tomorrow

DAY 2

08:30 - Coffee + Optional Q&A
09:00 - ggplot2
09:45 - Break
10:00 - ggplot2 Exercises
12:00 - Lunch
13:00 - Applied Statistics
14:30 - Break
14:45 - Data Analysis Exercise
15:45 - Cool things in R + Course Evaluation
16:00 - Bye Bye Bye!

— FROM EXCEL TO R

LET'S GET STARTED



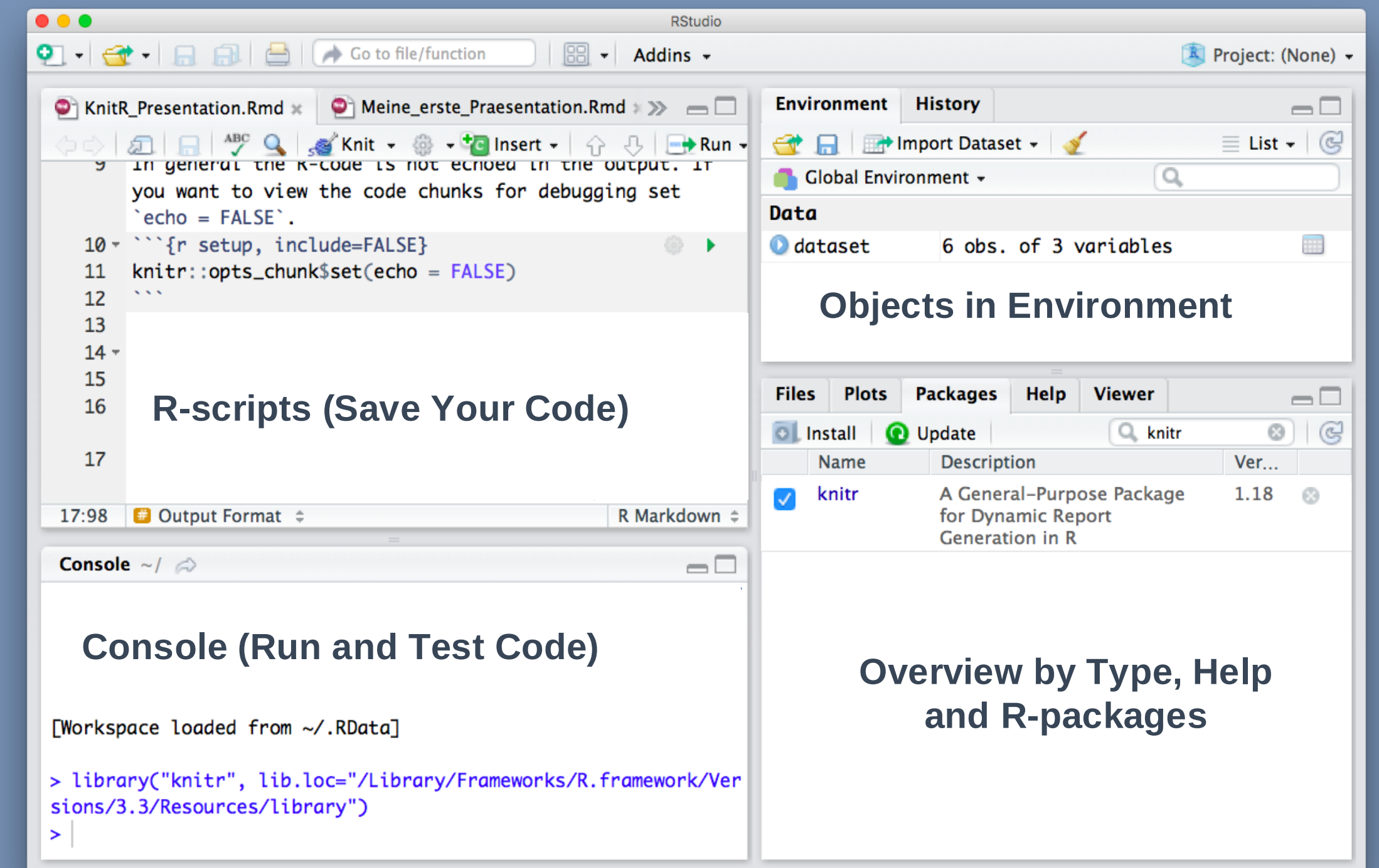
R & FRIENDS



Scripting / Programming Language



Reports (html, pdf, word)

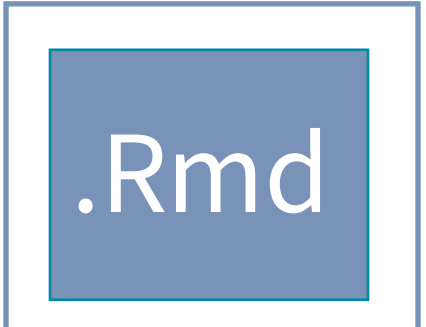


R Code Interpreter and Editor

FROM EXCEL TO R

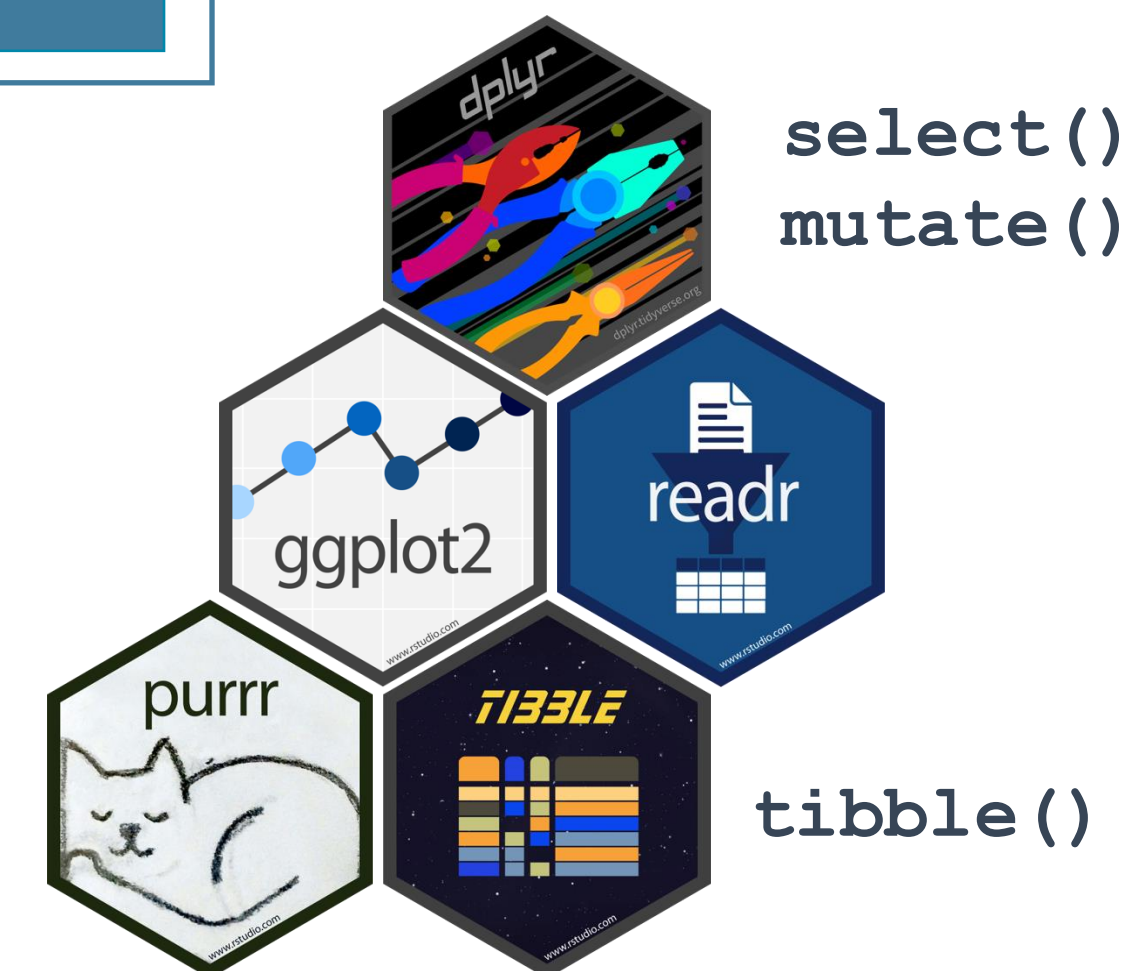
THE ANATOMY OF R

The **programming language**  may be used from within in the **Code Editor** 

The R code you write may be saved in **scripts** with extensions:   

Functions perform actions based on **input arguments** and return an output: `mean(x)`
`sd(x)`

Packages are bundles/libraries of functions.



QUARTO DOCUMENTS



- Tool which supports **reproducible** coding/publication (reports)
- Quarto works with a **markup language**
- A markup language is **text-coding system** which specifies structure & format of a document and the text within it
- We **render** (compile) a Quarto → html, pdf, word, website

Pointers for quarto:

- Separate sections with headers
- Do not overfill code chunks
- Write meaningful and precise comments

90% of all code comments:

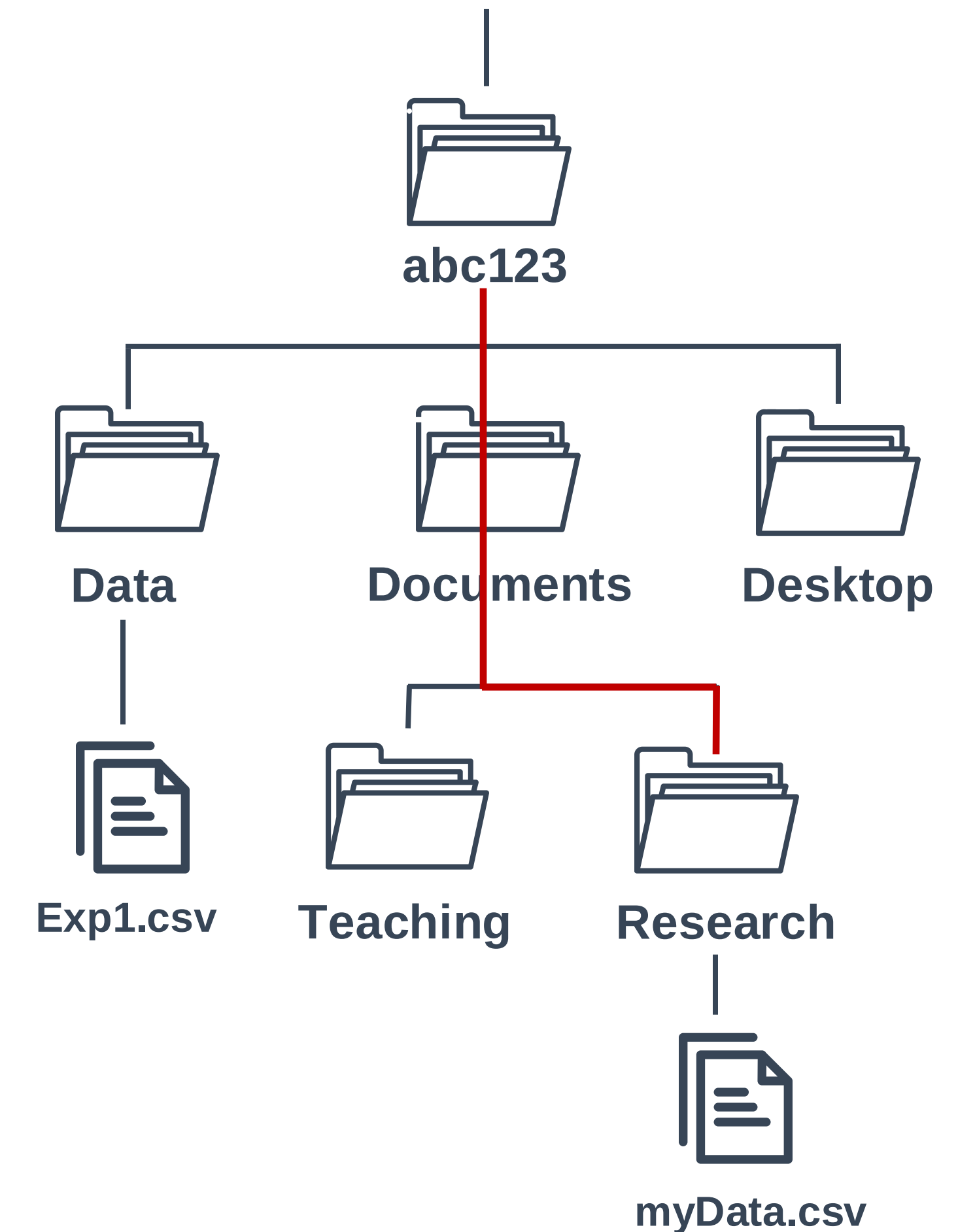


PATHS – LOCATION, LOCATION, LOCATION

- The directories on your computer are organized in a **File Tree**
- **Move** between directories or **point to** a place in the tree with a specific file

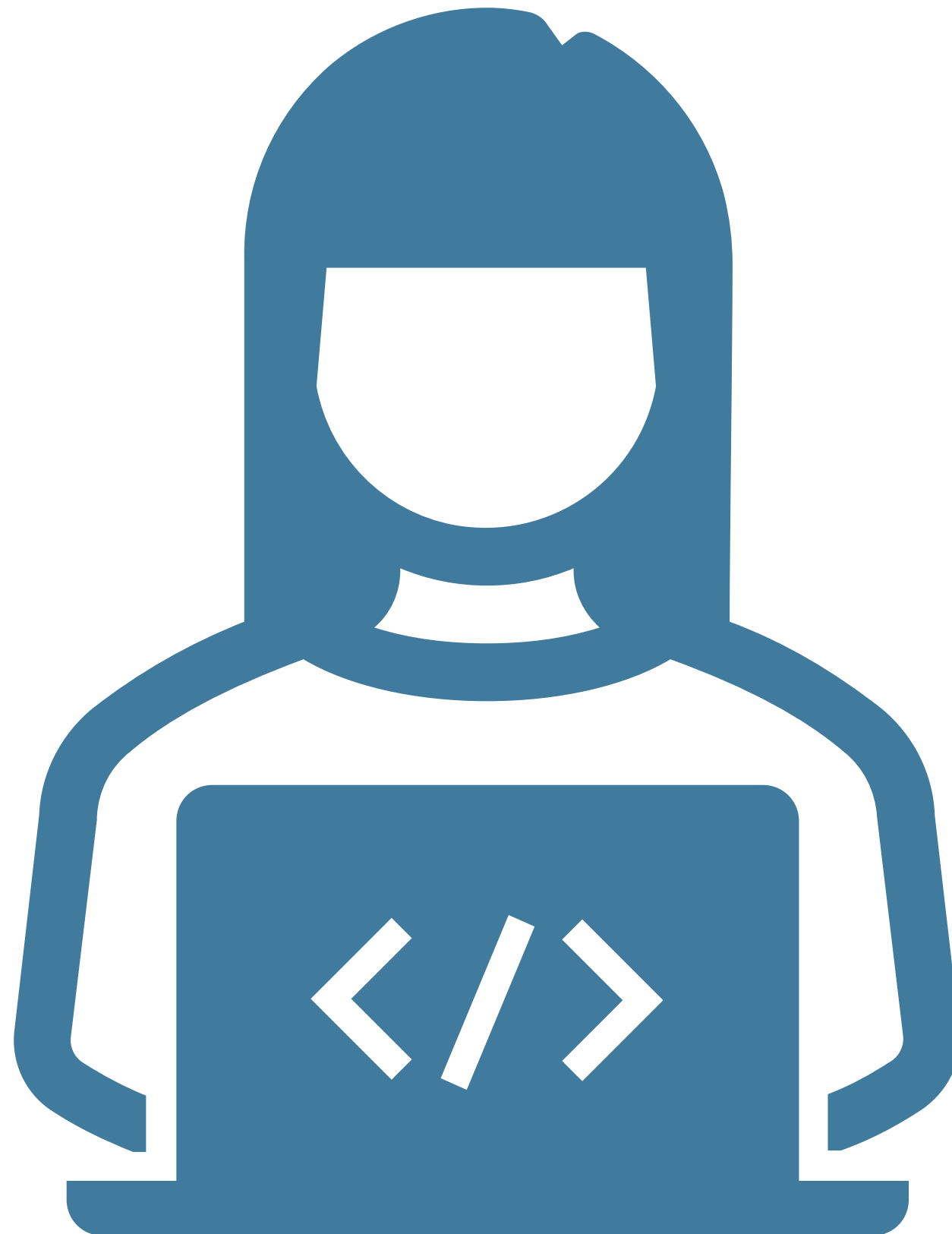
We do this by specifying a **path**:

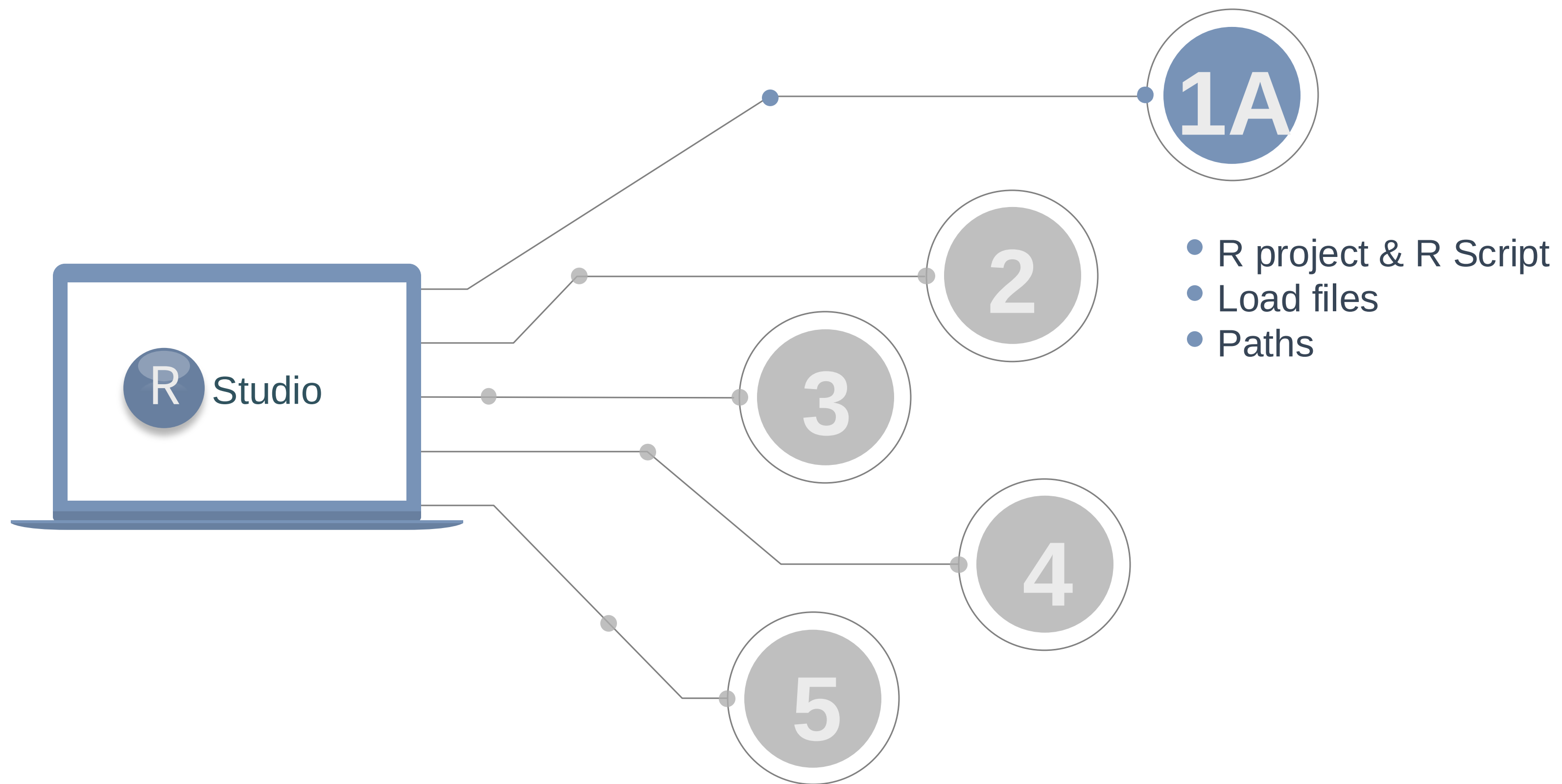
- **Move to:** `/abc123/Documents/Research`
 - **Point to:** `/abc123/Documents/Research/myData.csv`
-
- The place you are, right now, is called the **working directory** (wd)
Where am I: `getwd()`
Go to : `setwd('path')`



— FROM EXCEL TO R

LIVE CODING 1A – INTRO TO R





— FUNDAMENTALS

EXERCISE 1A

QUARTO CHEAT SHEET

YAML parameter:

```
---
title: My Project Name
output:
  html_document (pdf_document, ...)
---
```

Code Chunk:



Source mode:

```
```{r}
some R code
```
```

Visual mode:

```
{r}
# some R code
```

GETTING
STARTED

Code Options:

```
{r echo = FALSE} # don't print code (default is TRUE)
{r eval = FALSE} # don't run code (default is TRUE)
{r error = FALSE} # don't display error message (default is TRUE)
(Can also be set for warning and message)
```

Figure Options:

```
fig.align (= 'left', 'right', 'center')
fig.cap (= 'my figure caption')
fig.height (= n), fig.width (= n)
```

CHUNK
OPTIONS

Source mode:

Header

Header size ranging from largest (one #)
to smallest (six #):
my.text, ## my.text, ### my.text, etc.

Text

```
*italics*
**bold**
`highlighted`
```

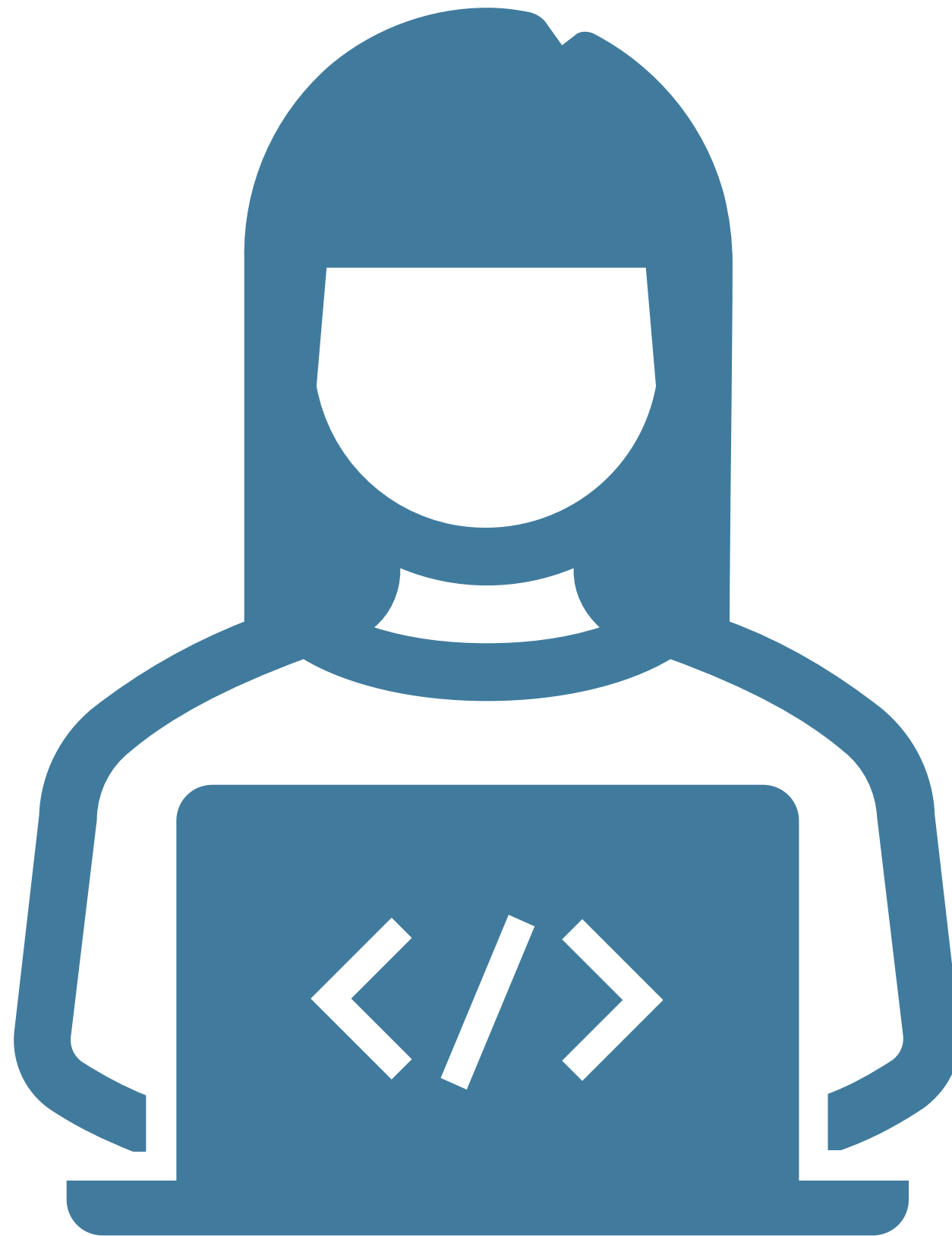
Lists

```
* List item1 (filled dot)
  + sub-item1 (open dot)
1. List item1 (numbered)
  i) sub-item1 (roman)
```

TEXT

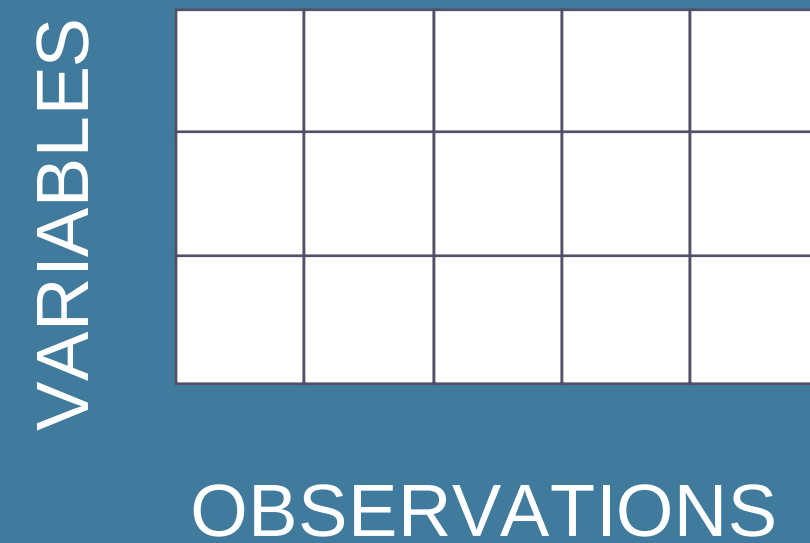
— FROM EXCEL TO R

LIVE CODING 1B – INTRO TO R

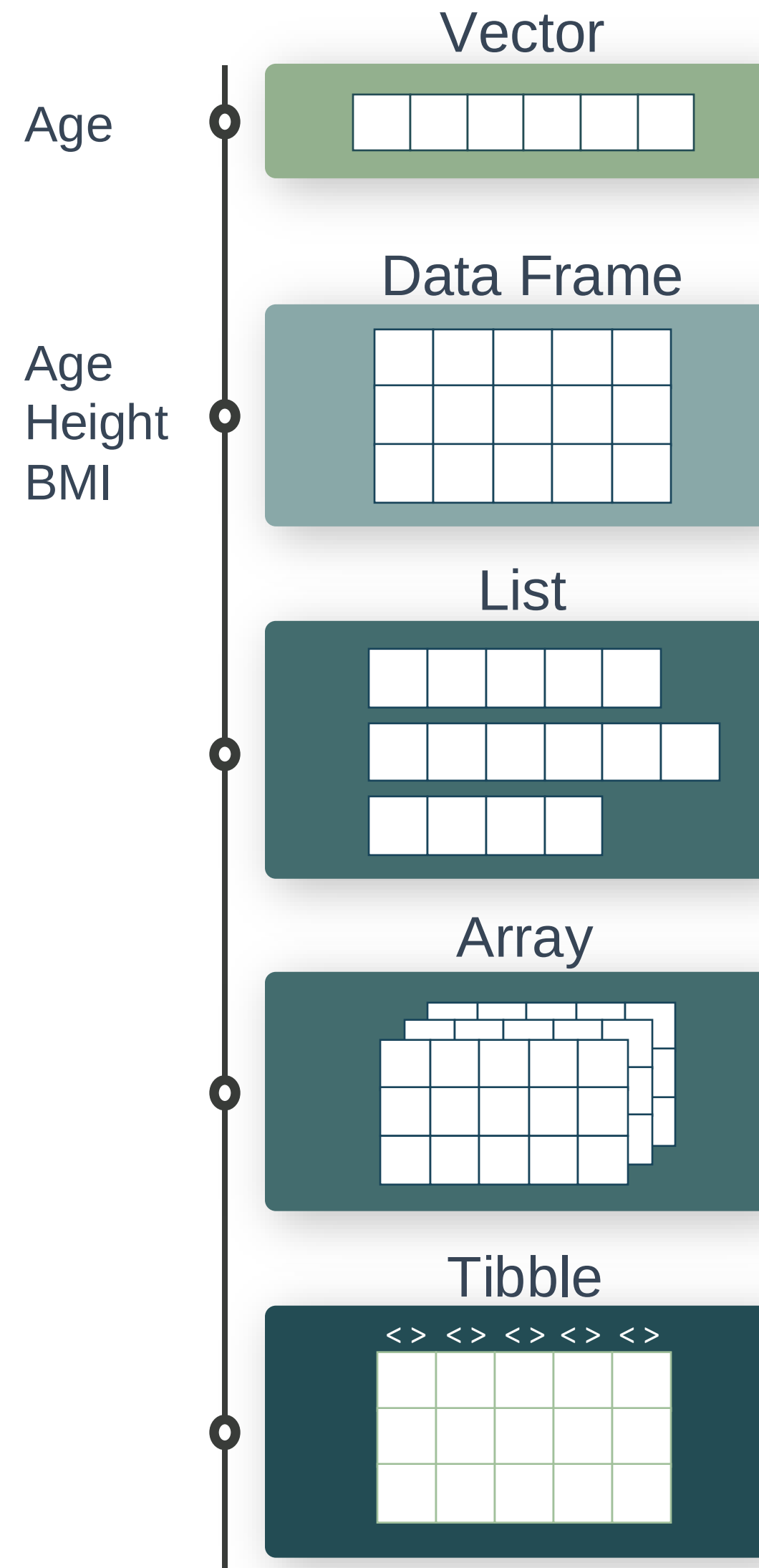


FROM EXCEL TO R

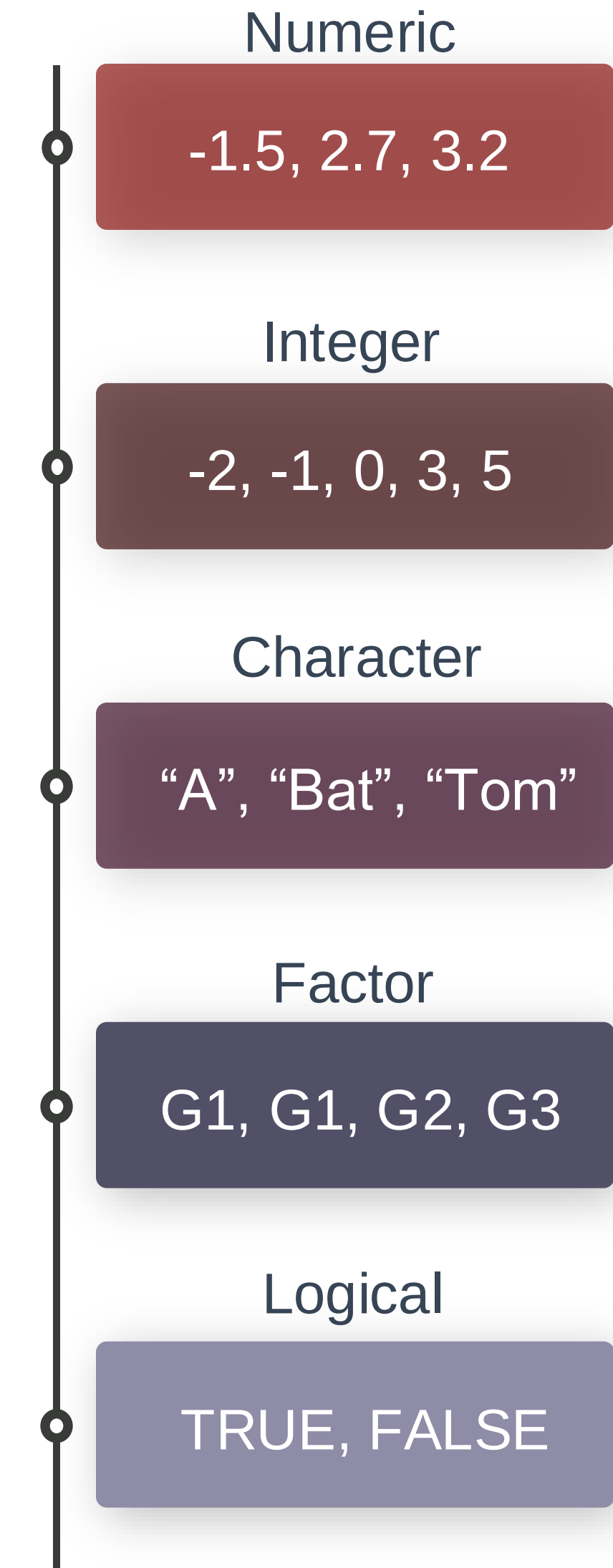
R DATA TYPES & STRUCTURES



DATA STRUCTURES



DATA TYPES



INTRO CHEAT SHEET

Basics:

```
getwd() # location
install.packages("package_name") # install packages
library(package_name) # load packages
```

Read in Data:

```
read.xlsx("name.xlsx") # library(readxl)
read.delim("name.txt", sep = "\t")
read.csv("name.csv", sep=";")
```

GETTING
STARTED

Overview:

```
head(df, n=10), tail(df, n=10) # first or last 10 rows
unique(), table(), count() # unique vals, count vals
```

```
view() # view data as table
df$col1 # extract column from dataframe
nrow(df), ncol(df) # number of rows/columns
```

OVERVIEW

Type/Class:

```
class() # get data type/class
is.numeric(x), is.character(), is.factor() # is data type
as.numeric(x), as.character(), as.factor() # change data type
```

Conditions:

```
a == b    a != b    a > b    a < b
a >= b    a <= b    is.na() is.null()
```

DATA TYPES &
CONDITIONS

Summary Stats:

```
summary() # summary statistics
mean(), median(), sd(), sum(), log(), max(),
min(), quantile(), var()
```

Plots:

```
plot(x)
plot(x, y) or plot(col1, col2, df) # scatter
hist(x) # histogram
```

STATISTICS &
BASE PLOTS

ONLINE RESOURCES FOR R

<https://www.r-project.org/>



GET STARTED

<https://rseek.org/>

<https://rstudio.com/resources/cheatsheets/>

<http://www.cookbook-r.com/>

<https://www.statmethods.net/r-tutorial/index.html>



GRAPHICS

<https://www.r-graph-gallery.com/>

<http://r-statistics.co/Top50-Ggplot2-Visualizations-MasterList-R-Code.html>



BOOKS & COURSES

<https://www.r-bloggers.com/best-books-to-learn-r-programming/>

<https://www.datacamp.com/>

<https://www.codecademy.com/>

<https://www.coursera.org/>



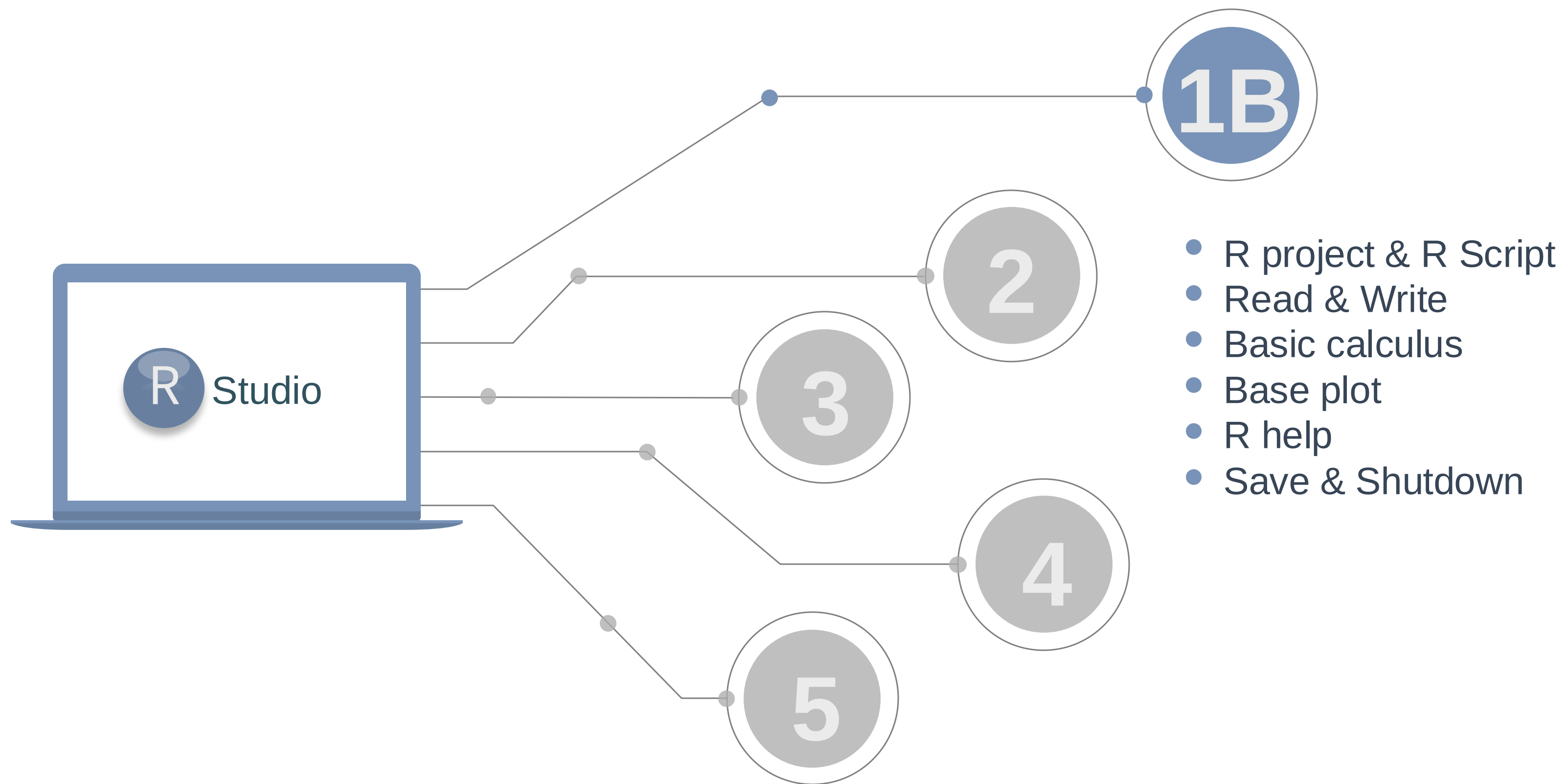
OTHER RESOURCES

<https://github.com/trending/r>

<https://blog.revolutionanalytics.com/>

<https://stackoverflow.com/questions/tagged/r>





— FUNDAMENTALS

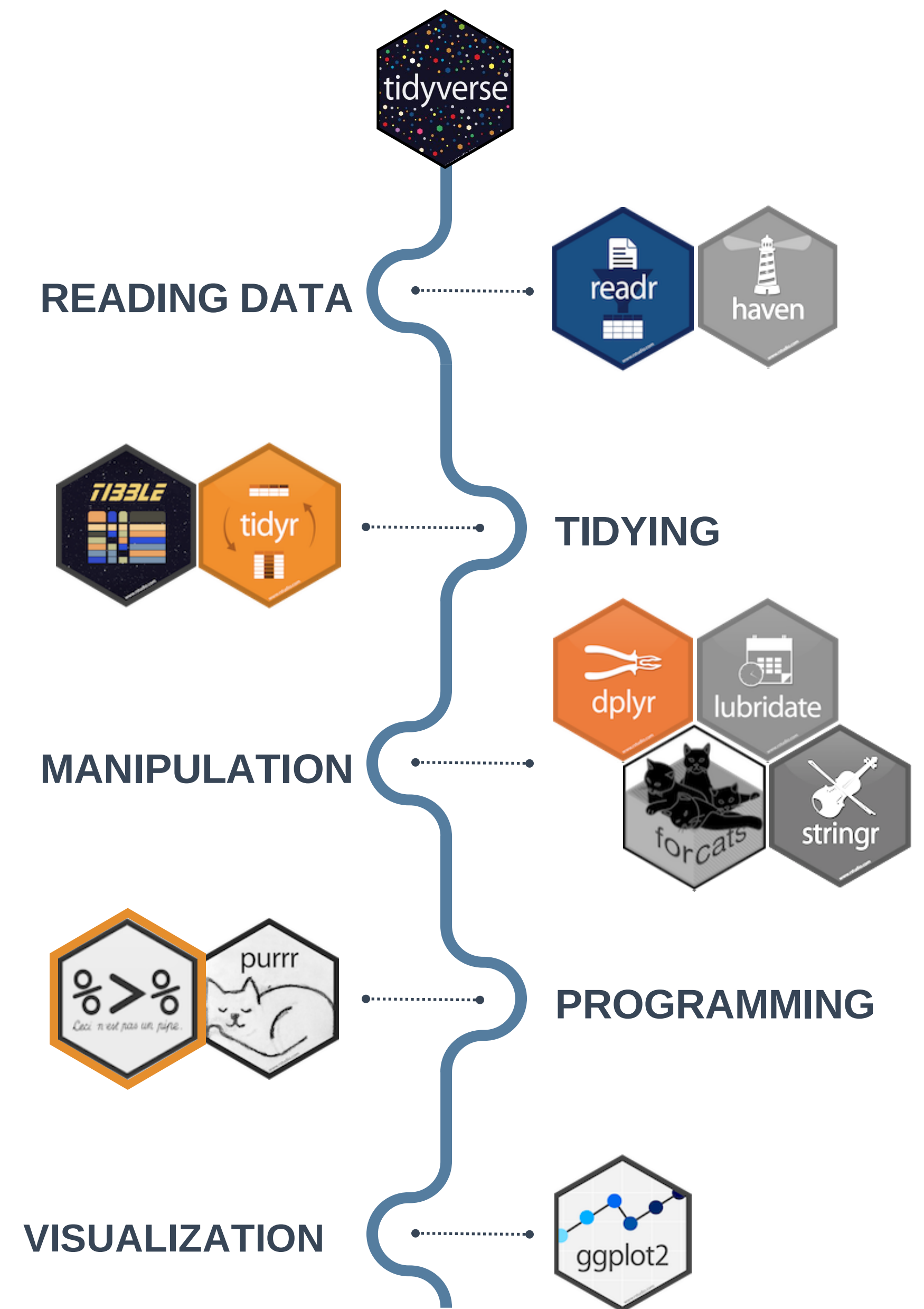
EXERCISE 1B

<https://www.tidyverse.org/>
TIDYVERSE

tidyverse is a collection of R packages for data science

“The packages share an underlying design philosophy, grammar, and data structures.” *Wickham and Grolemund*

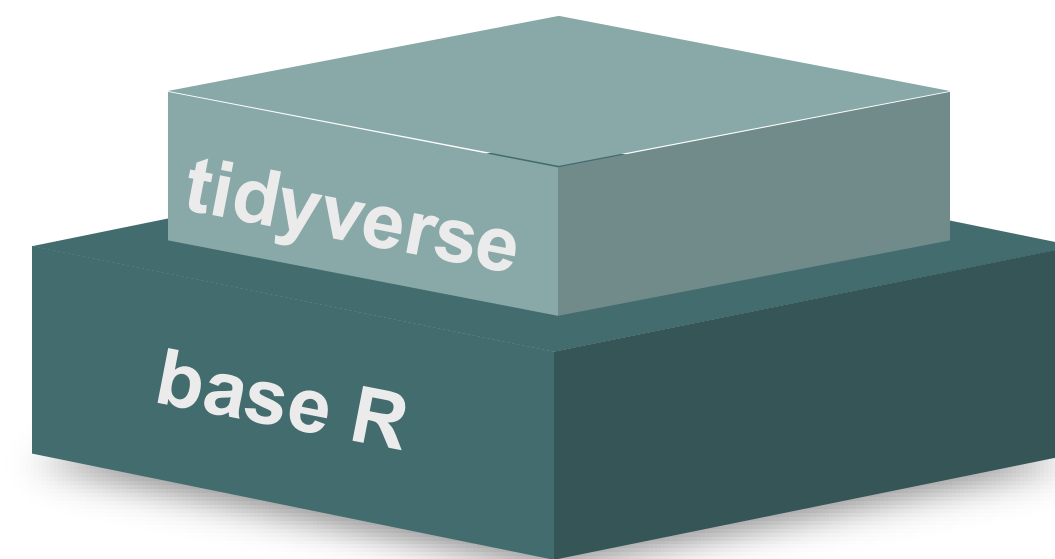
tidyverse is used to “tidy up” your datasets, so they are easy to work with



CECI N'EST PAS UNE PIPE

%>%

- You do NOT have to “chose” between tidyverse and base R



BENEFITS

- Short & well-organised code
- Tidy datasets, easy to work with
- Great documentation
- Functions with logical names & inputs

CONSIDERATIONS

- Can be less stable
- “Different syntax”
- Can be computationally slow

base R

```
# think from the inside out  
g(f(x,y),z)
```

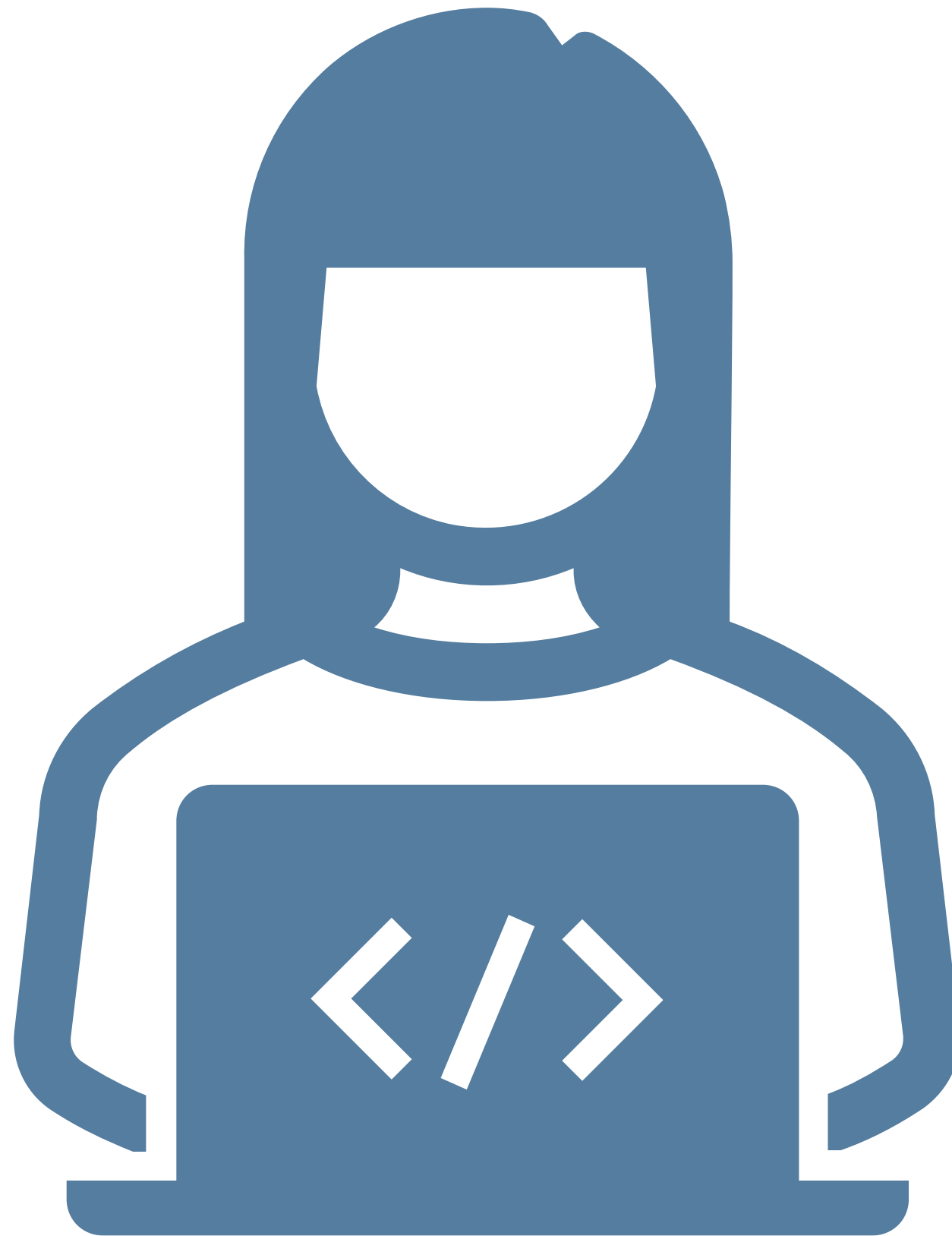
tidyverse

```
# no brain acrobatics  
x %>% f(y) %>% g(z)
```

↑ pipe symbol

— FROM EXCEL TO R

LIVE CODING 2 – TIDYVERSE



TIDYVERSE CHEAT SHEET

readr, tidyr, dplyr, ...

Read Data (*readr*)

Reading tabular data

There are solutions for multiple data types
`read_excel()` # using *readxl* package
`read_table()`
`read_csv()`

Useful arguments

Skip lines: `read_csv(file, skip=1)`
Read subset: `read_csv(file, n_max=1)`

Data types

`readr` guesses the types of each column and tells you about it
(“Parsed with column specifications: ...”)

Workflow

Tidyverse workflow

```
df %>%  
  select(col1)  
  
df %>%  
  filter(col1 < x)  
  
df %>%  
  summarize(n_1 = n(),  
            avg_1 = mean(col),  
            sd_1 = sd(col1))
```

Select column

```
df$col1
```

Data Manipulation (*dplyr*)

Summary

```
summarize()  
count()
```

Group

```
group_by()  
Functions will manipulate each group separately and combine results.
```

Extract and sort observations (rows)

```
filter() # subset rows by condition  
distinct() # subset to unique values  
top_n() # subset by position  
arrange() # sort low->high, other way with desc()
```

Manipulate variables (columns)

```
select() # subset rows by condition  
mutate(new_name = f(column))  
mutate(new_name = ifelse(col1 < x, "Yes", "No"))  
mutate(new_name = col1 + col2)
```

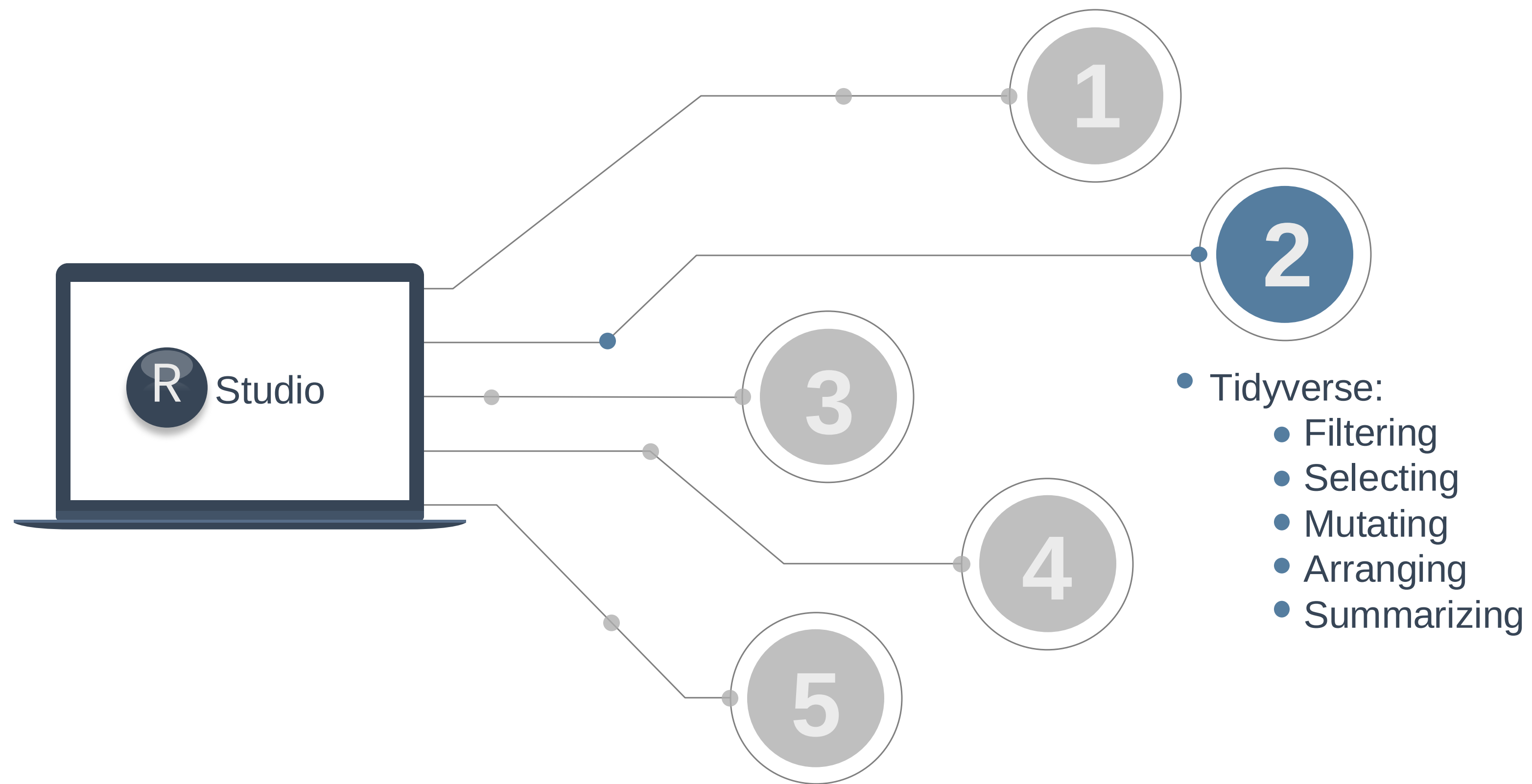
HELP

R Documentation (e.g. enter `?dplyr::filter` and see examples)

Much more info and detailed cheat sheets:

<https://brianward1428.medium.com/introduction-to-tidyverse-7b3dbf2337d5>

It also helps to google “tidyverse + whatever_you_want_to_do”



— TIDYVERSE EXERCISE 2

GGPLOT2 - EASY GRAPHICS



Aesthetically pleasing graphics.



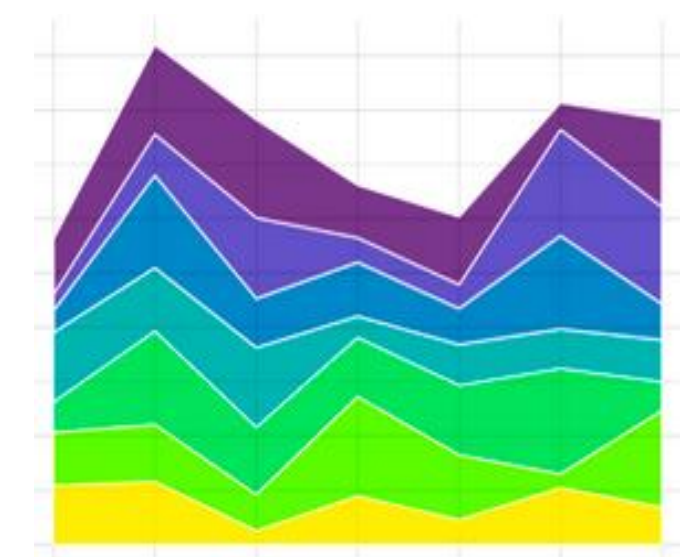
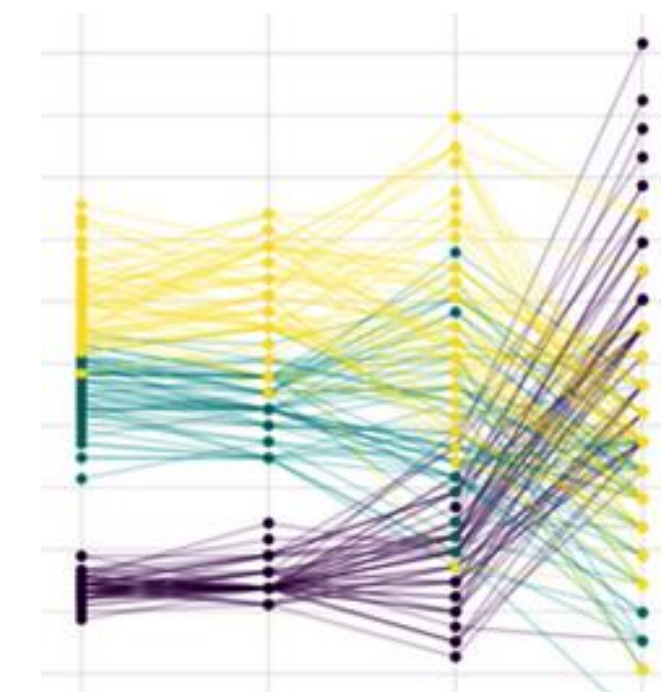
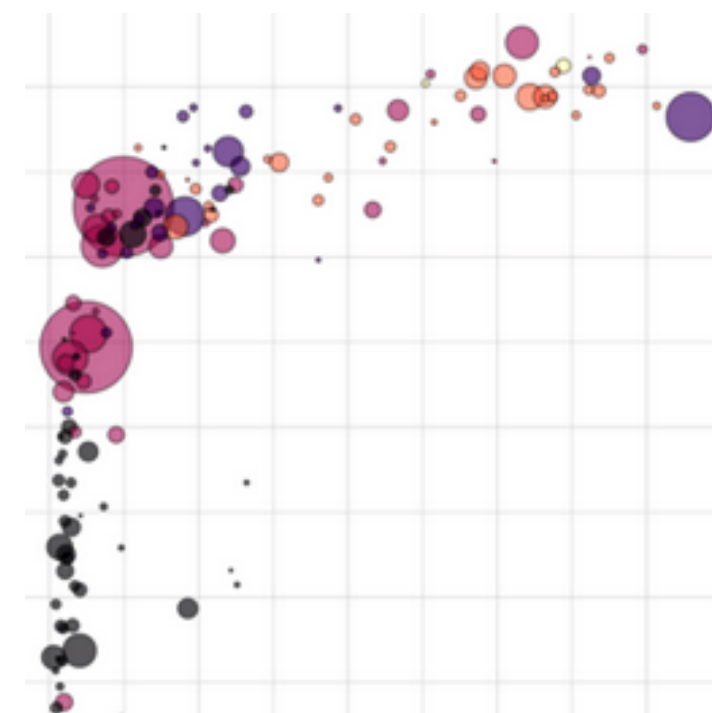
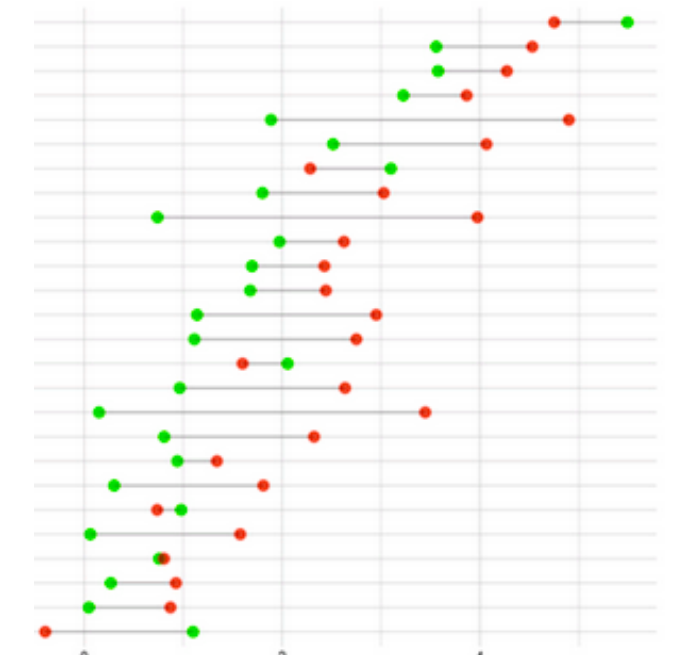
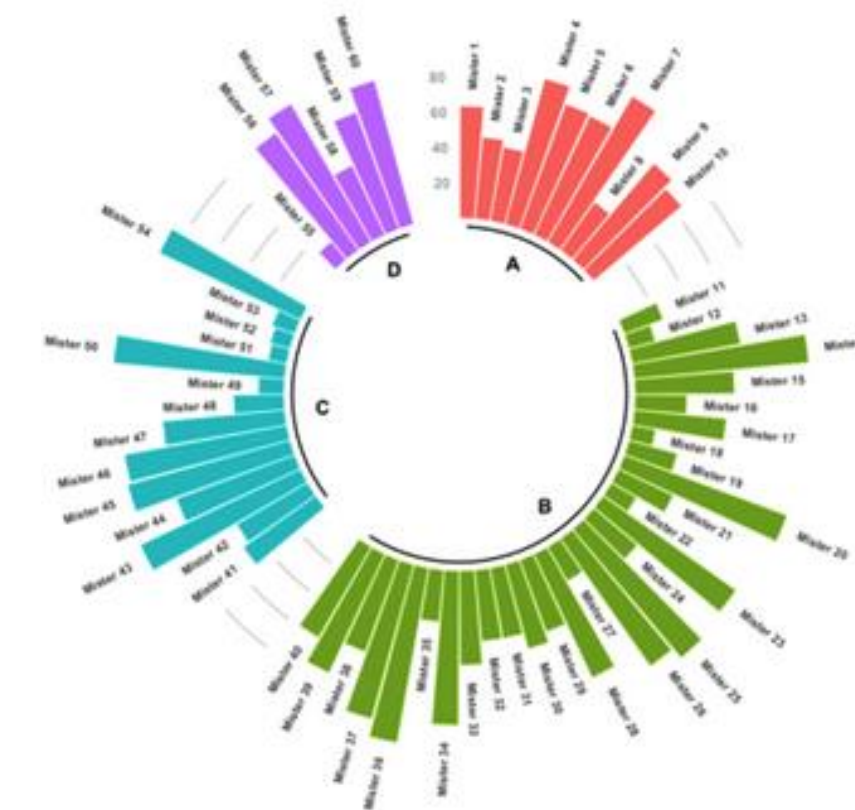
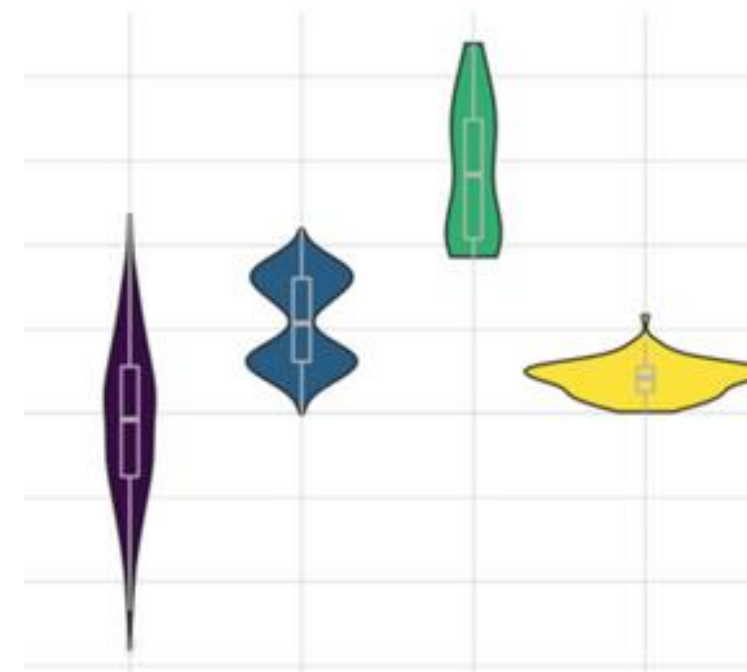
Well-defined “additive” (+) structure.



Integrates perfectly with tidy data.



Great documentation & community



GGPLOT2

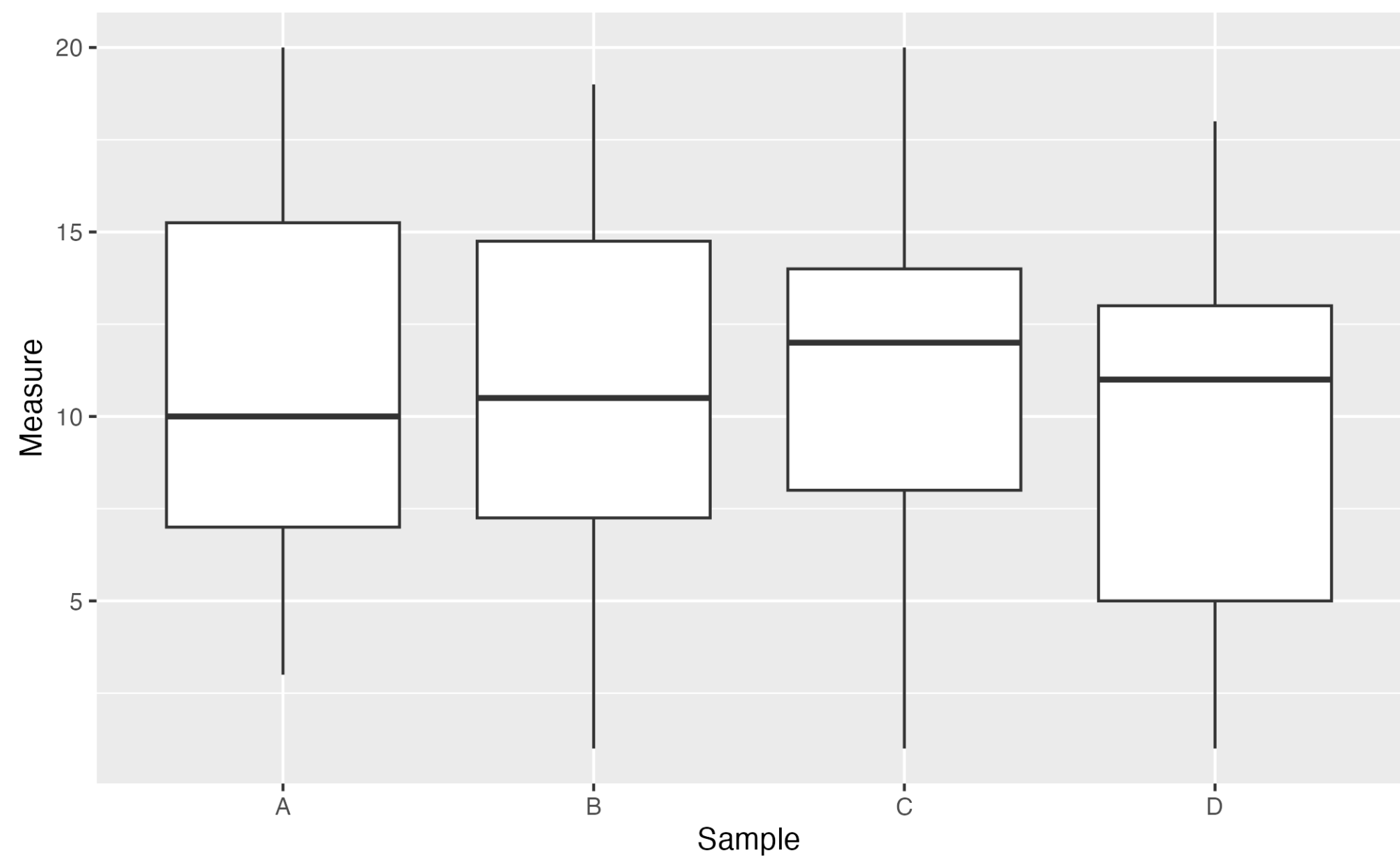
ADDITIVE STRUCTURE

DATASET, SAMPLES
& OBSERVATIONS

DEFINE PLOT TYPE

```
ggplot(df,  
  aes(x = Sample, y = Measure))
```

```
ggplot(df,  
  aes(x = Sample, y = Measure)) +  
  geom_boxplot()
```



head(df)

...

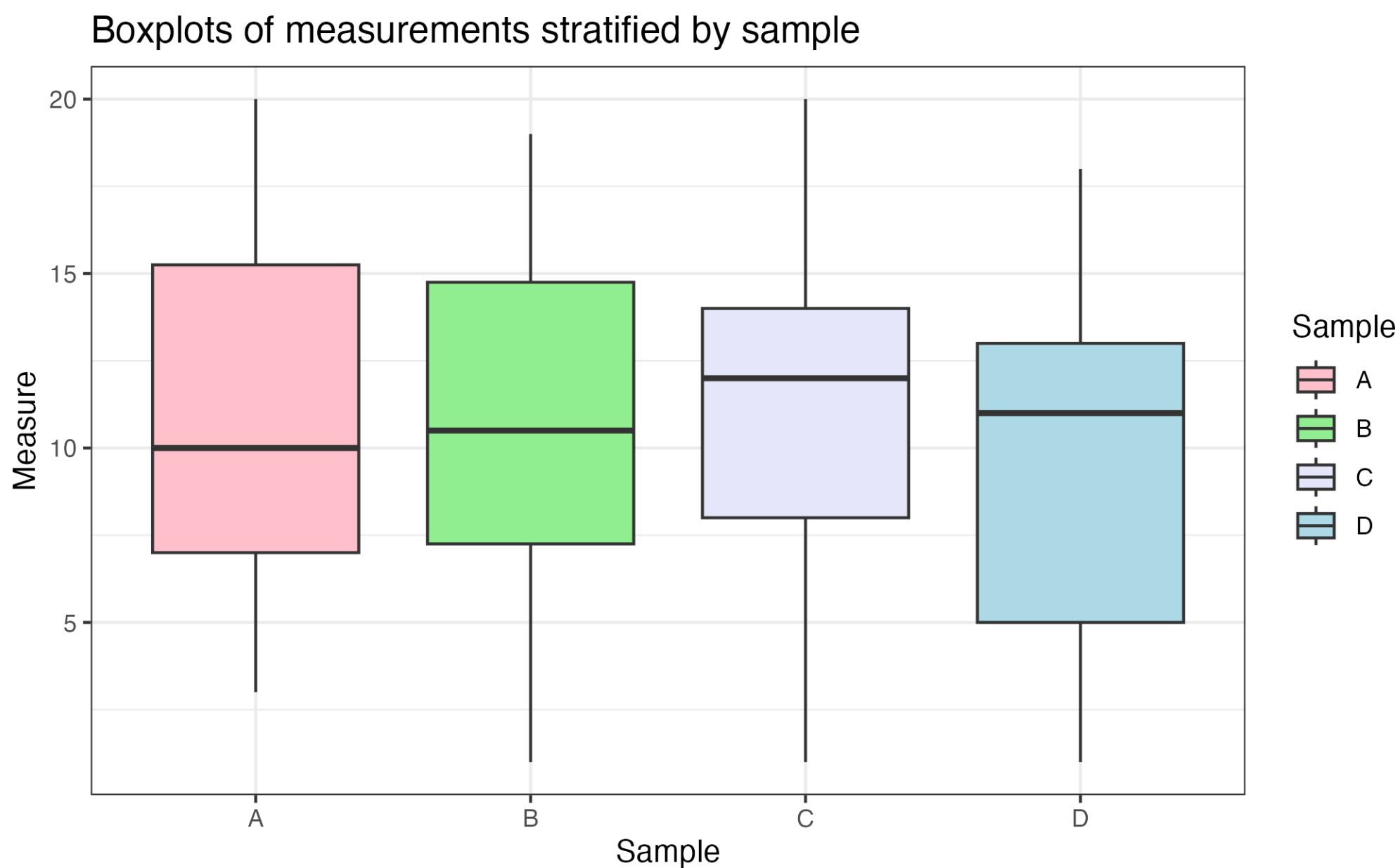
Description: df [6 × 2]

| | Sample<chr> | Measure<int> |
|---|-------------|--------------|
| 1 | C | 20 |
| 2 | C | 14 |
| 3 | C | 3 |
| 4 | B | 8 |
| 5 | C | 16 |
| 6 | B | 12 |

6 rows

GGPLOT2

ADDITIVE STRUCTURE



DATASET, SAMPLES
& OBSERVATIONS

DEFINE PLOT TYPE

COLOR BY GROUP

TITLE AND LEGEND

CUSTOM COLORS

BACKGROUND

```
ggplot(df,  
  aes(x = Sample, y = Measure))
```

```
ggplot(df,  
  aes(x = Sample, y = Measure)) +  
  geom_boxplot()
```

```
ggplot(df,  
  aes(x = Sample, y = Measure,  
    fill = Sample)) +  
  geom_boxplot()
```

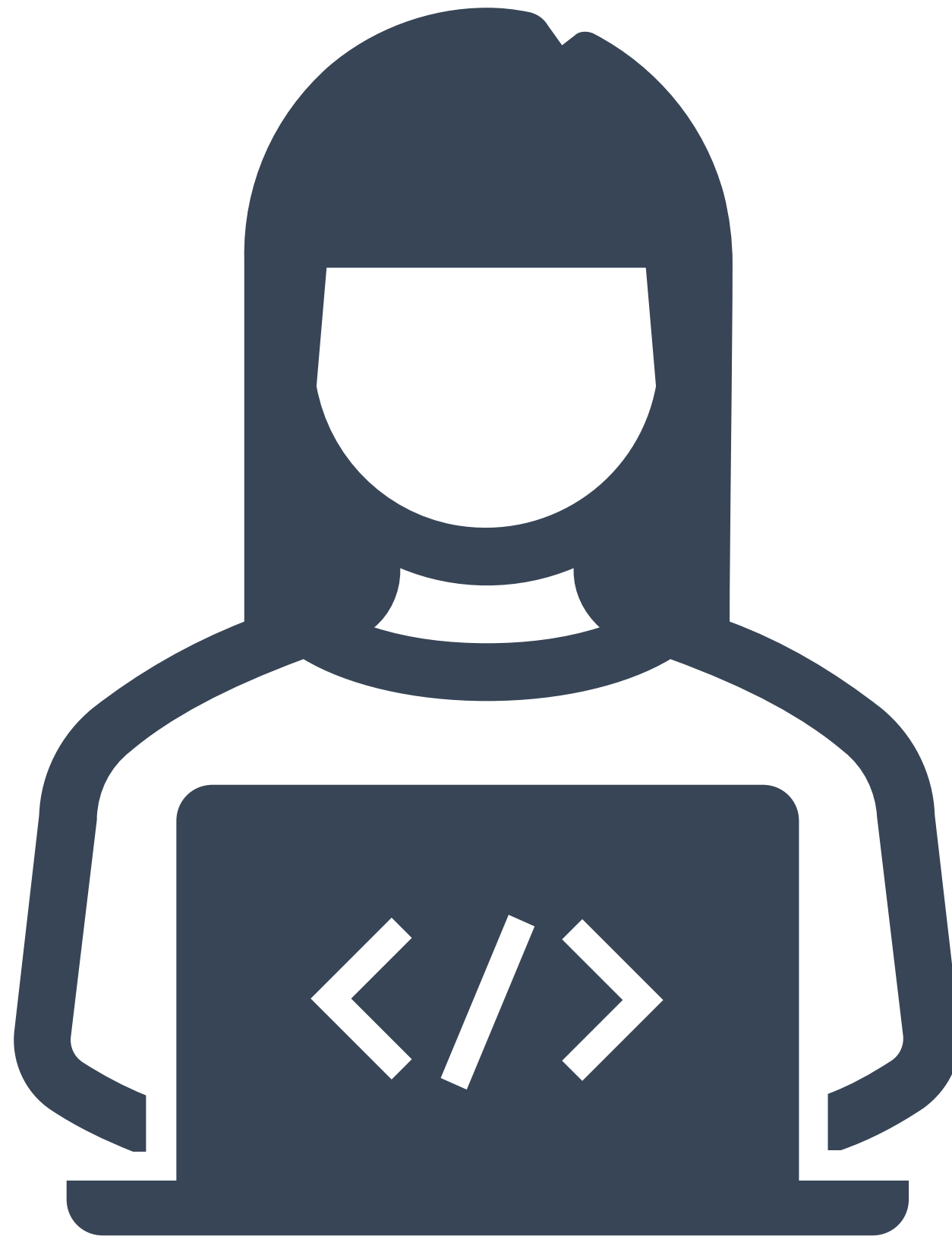
```
... +  
  labs(title = "Boxplots of  
measurements stratified by sample")
```

```
... +  
  scale_fill_manual(values =  
c("pink", "lightgreen", "lavender",  
"lightblue"))
```

```
... +  
  theme_bw()
```

— FROM EXCEL TO R

LIVE CODING 3 – GGPLOT 2



GGPLOT CHEAT SHEET

Define Plot:

```
ggplot(data = my.data,  
       aes(x = x.var,  
           y = y.var))
```

Add Plot Type:

```
... + geom_point() # scatter plot  
... + geom_line()  
... + geom_boxplot()  
... + geom_col()  
... + geom_density()  
... + geom_histogram()
```

GET
STARTED

One Color:

```
ggplot(..., aes(...),  
       color = "green")
```

Color Fill by Group:

```
ggplot(..., aes(...,  
               fill = z.var))
```

Custom Colors:

```
... + scale_*_manual(values = c())  
ex: scale_color_manual(values = c("blue", "pink"))
```

COLORS

More Colors:

```
... + scale_fill_grey(start = 0.2, end = 0.8)  
... + scale_fill_gradient(low="white", high="red")
```

Labels:

```
... + labs(title = "Title",  
           x = "X label",  
           y = "Y label")
```

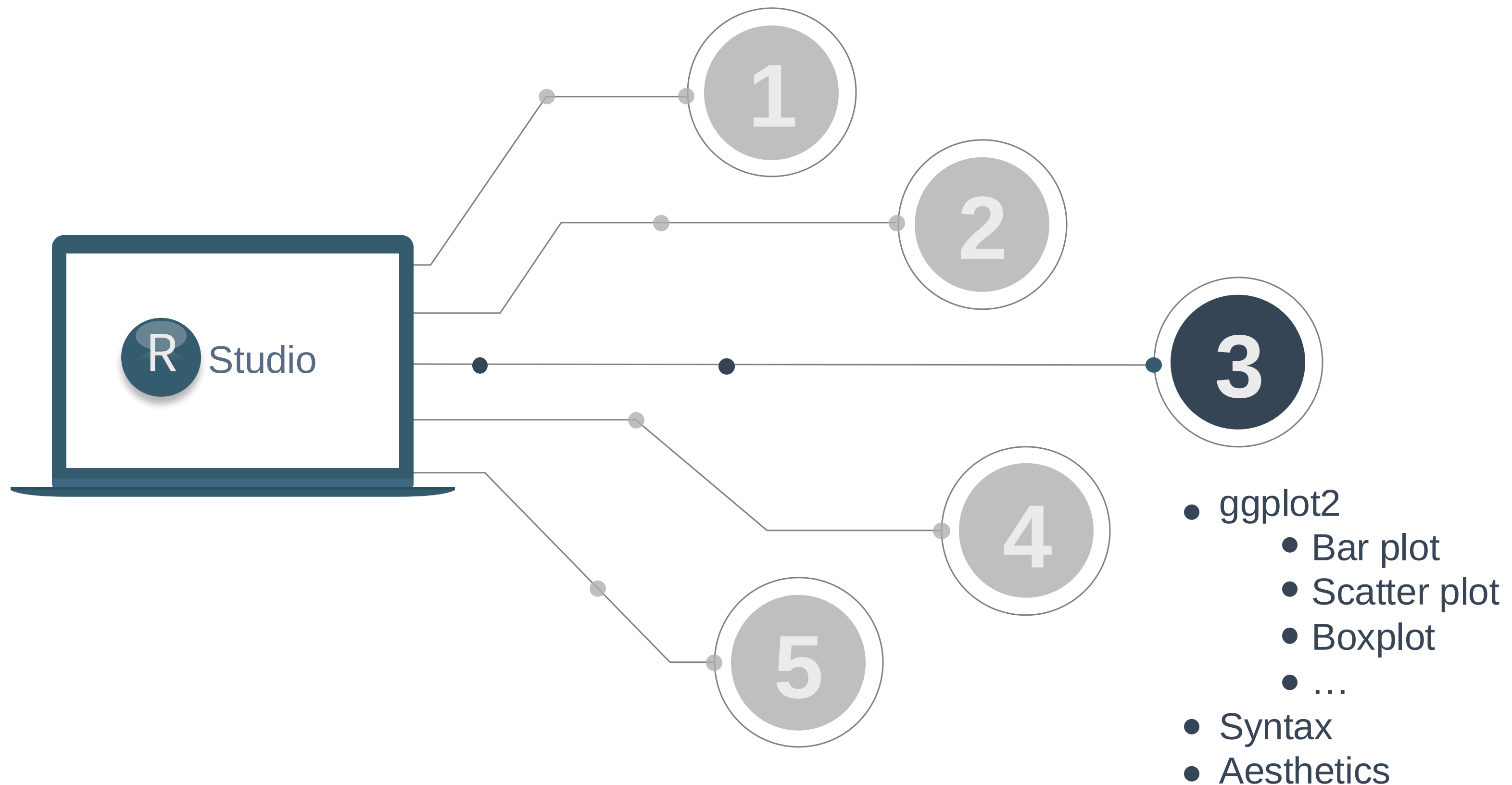
COLOR SCALES
& THEMES

Theme: ... + theme_bw()
... + theme_minimal()
... + theme_dark()
... + theme_classic()

Text:

```
... + theme(legend.position = *)  
* = "none", "top", "bottom", "left", "right"
```

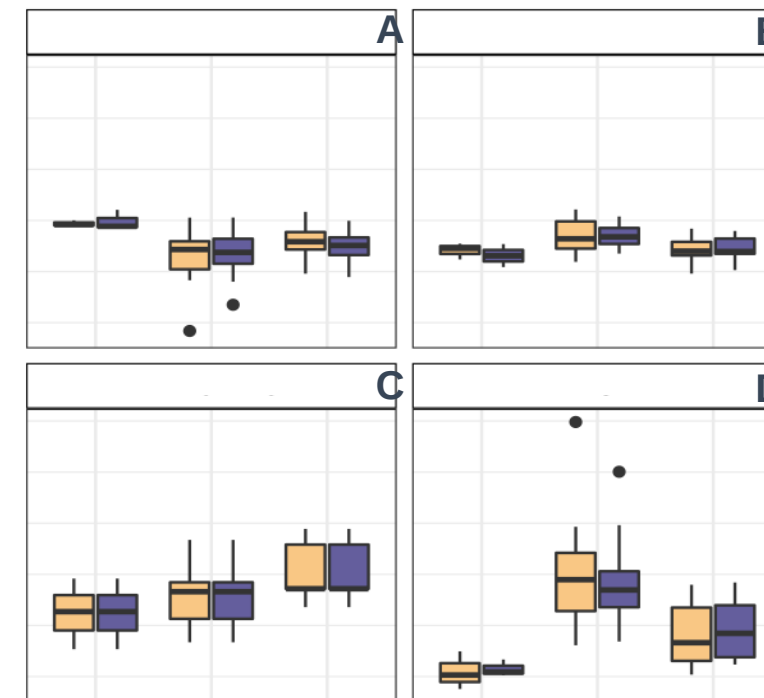
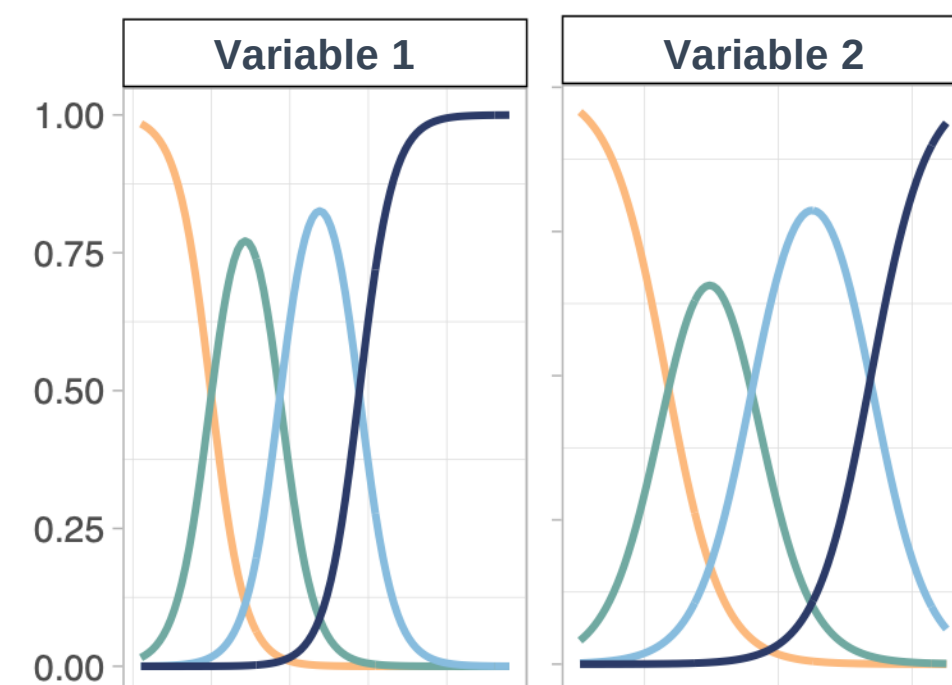
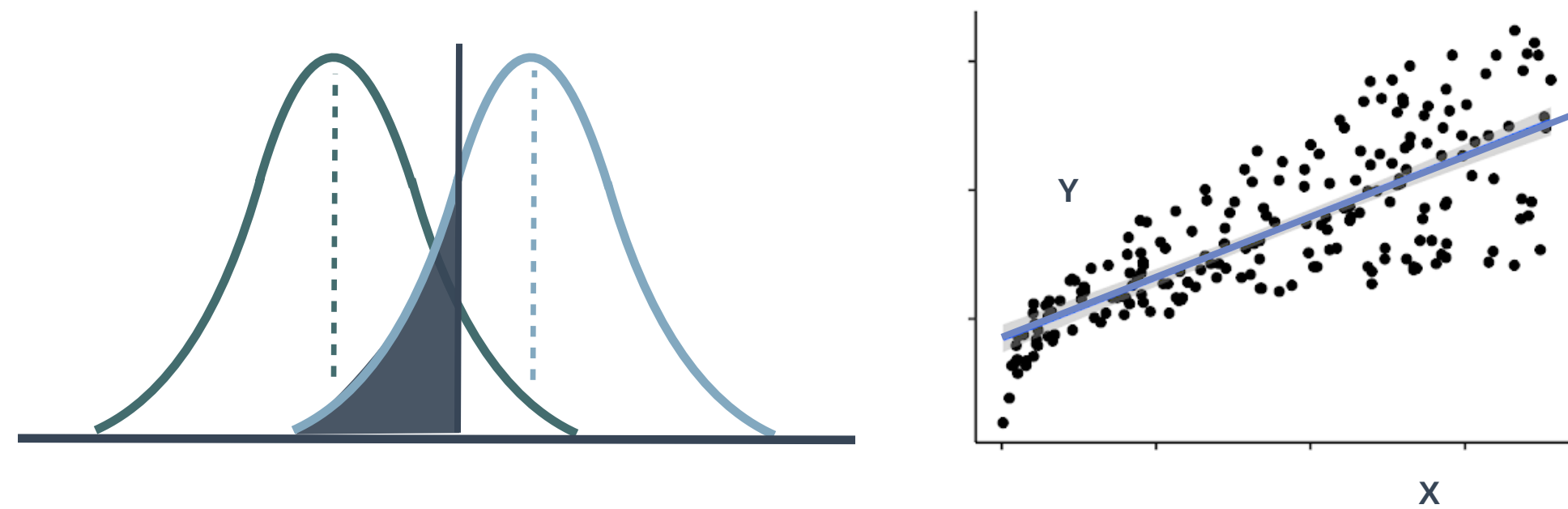
TEXT



GGPLOT2 EXERCISE 3

<http://www.sthda.com/english/>

R - A STATISTICAL SCRIPTING LANGUAGE



MODEL FUNCTIONS

```
lm(), glm()  
lmer(), glmer(),  
nls(), ...
```

EMMEANS PACKAGE

```
emmeans(),  
pairs(), cld()
```

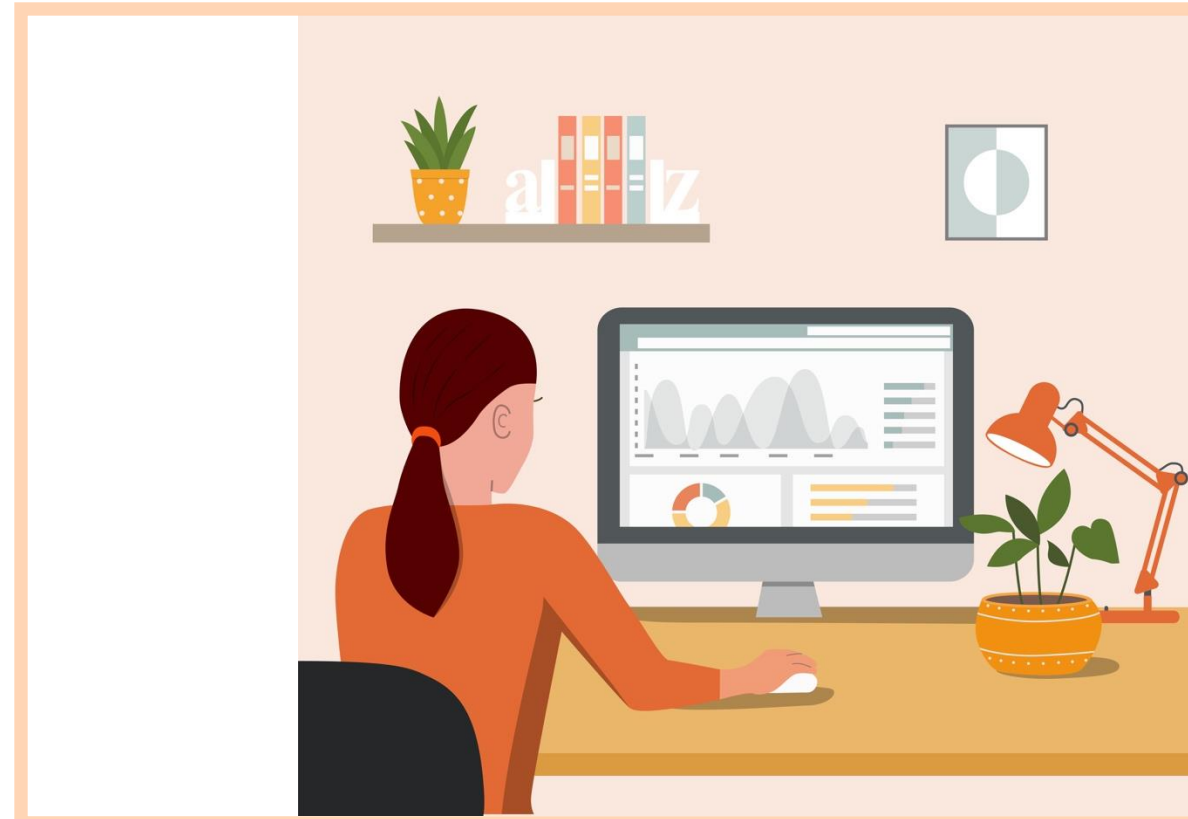
APPLY TO MODEL

```
summary(), anova(),  
confint(), predict(),  
drop1(), update(),  
step(), ...
```

MORE FUNCTIONS

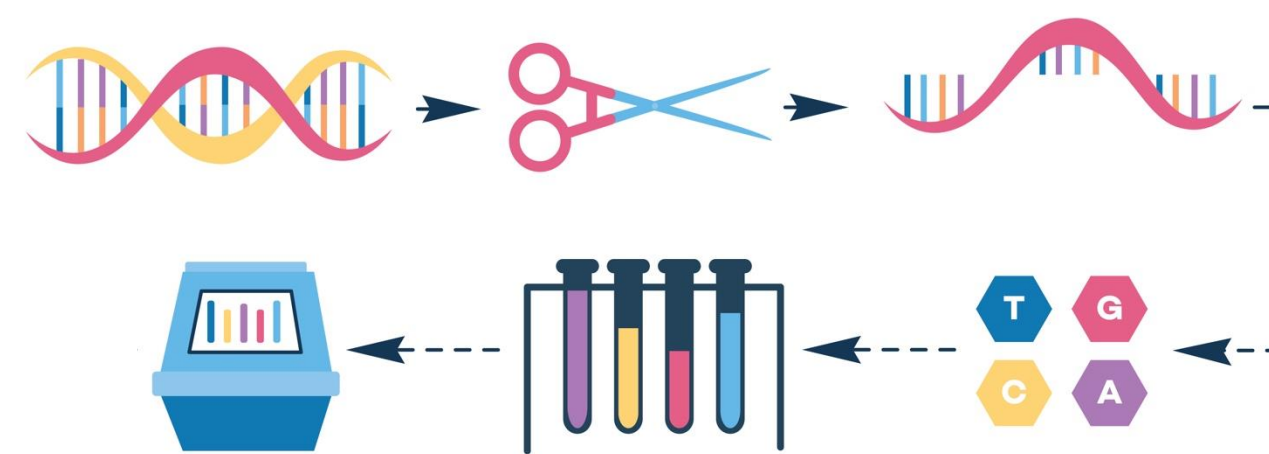
```
t.test(), cor(),  
cor.test(), aov(),  
quantile(),  
p.adjust(),  
rank(), ...
```

Let's use R in a statistical analysis



YOU

The researcher with R skills!



YOUR DATA

Gene expression data, i.e. how much a gene is used to produce a protein. Multiple genes measured in one experiment.



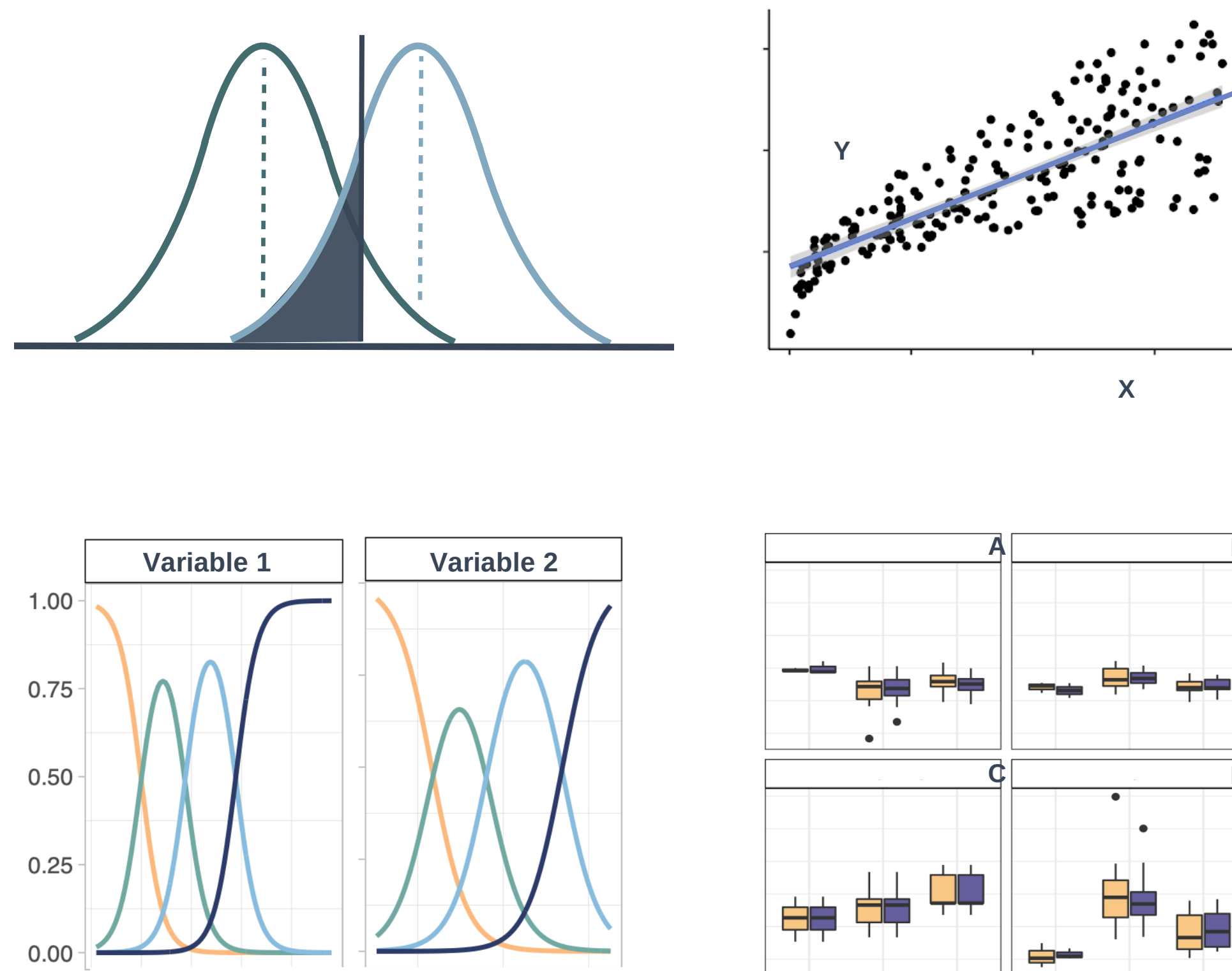
YOUR DISEASE OF INTEREST

Psoriasis, an immune-mediated skin disorder.

R - A STATISTICAL SCRIPTING LANGUAGE

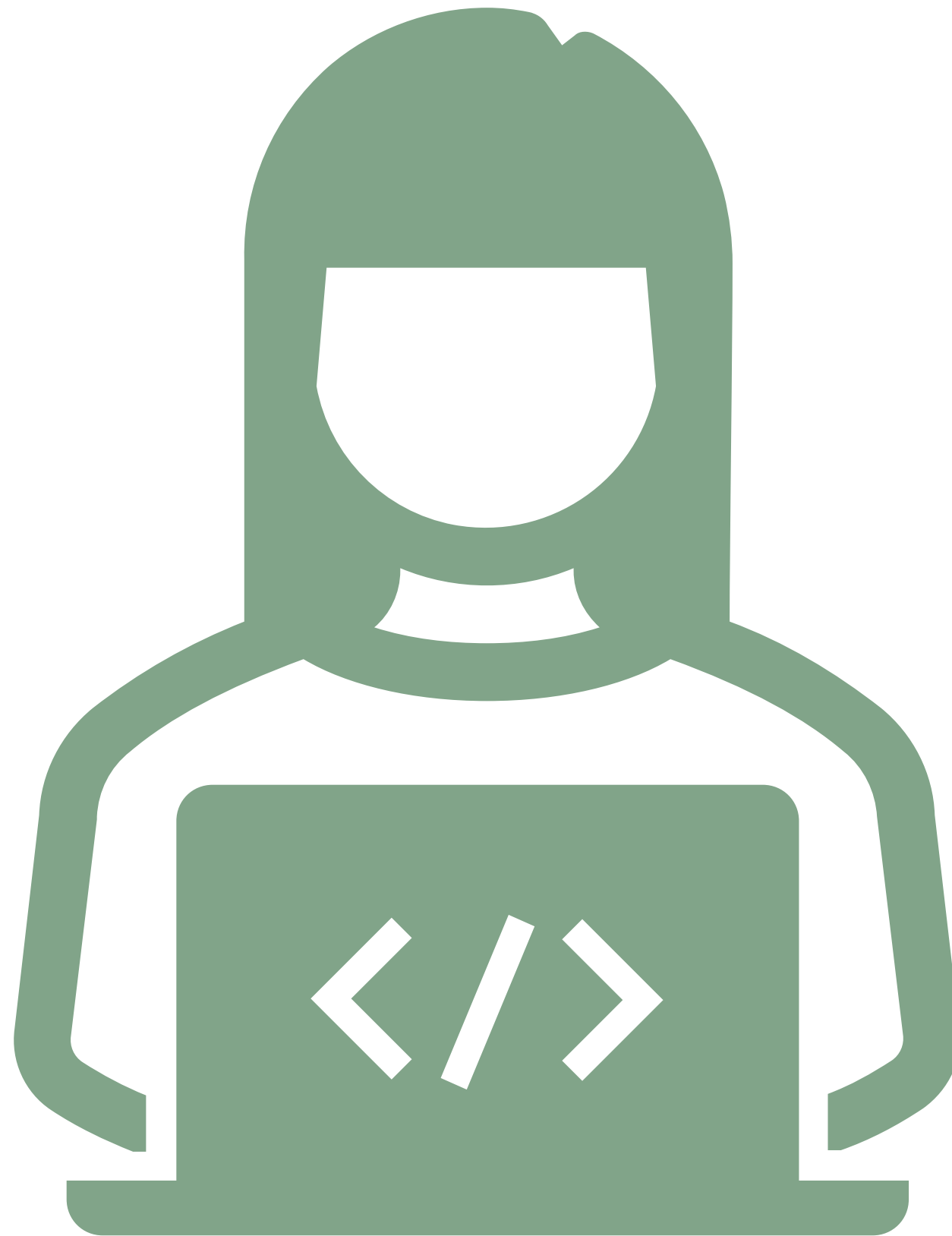
During this session:

- *Cooperatively* discuss and share ideas about the data
- Apply steps of basic statistical analysis for hypothesis testing consistent with the given data
- Suggest conclusions based on your analysis, regarding the association between psoriasis and gene expression levels



— FROM EXCEL TO R

LIVE CODING 4 – APPLIED STATS



<https://rstudio.com/resources/cheatsheets/> for various relevant cheat sheets.
Other example: <https://www.dummies.com/programming/r/statistical-analysis-with-r-for-dummies-cheat-sheet/>

STATS CHEAT SHEET

Import Data:

```
read_excel("my.data.xlsx")
```

Overview of Data:

```
summary(my.data)      length(my.data)  
nrow(my.data)         names(my.data)
```

GET
STARTED

Linear:

```
lm(y~x, data=my.data)  
confint(model)
```

Logistic:

```
glm(y~x,  
data=my.data)
```

Linear Mixed:

```
lmer(y~x + (1|z),  
data=my.data)
```

Check Model:

```
summary(model)  
par(mfrow=c(2,2))  
plot(model)
```

REGRESSION
MODELS

ANOVA:

```
anova(model2, model1)
```

F-Test:

```
drop1(model, test="F")
```

Emmeans:

```
emmeans(model, ~x)  
pairs(emmeans(model, ~x))
```

TESTS/COMPARI
SONS

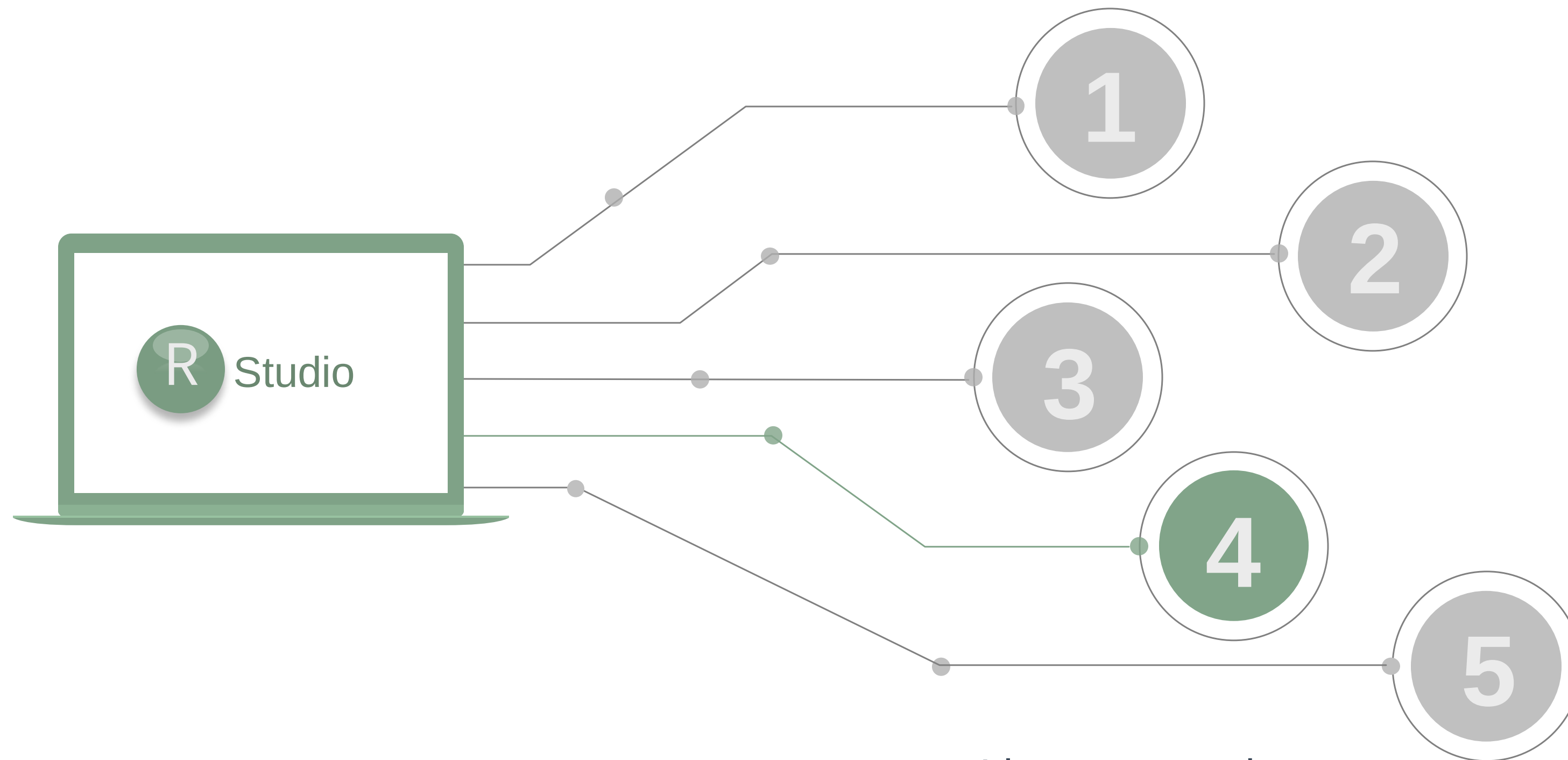
Check Type:

```
table(my.data$x)  
is.numeric(my.data$x)  
is.factor(my.data$x)
```

Change Type:

```
my.data <- mutate(my.data, x = factor(x))  
my.data$z <- as.numeric(my.data$z)
```

VARIABLE
S



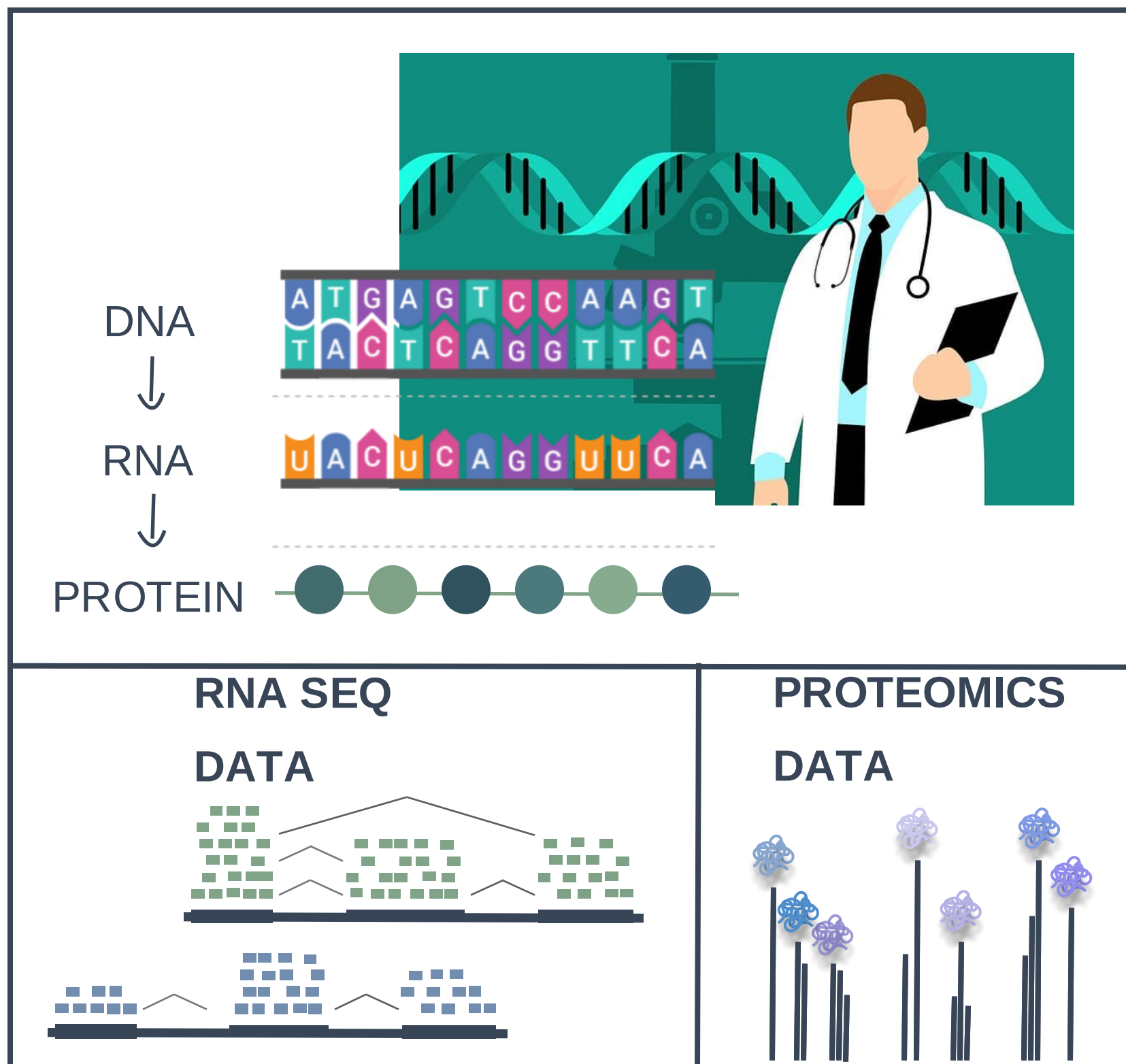
- Linear regression
- Summary Statistics
- ANOVA
- Logistic regression
- Clustering
- Correlation

Statistics in R

EXERCISE 4

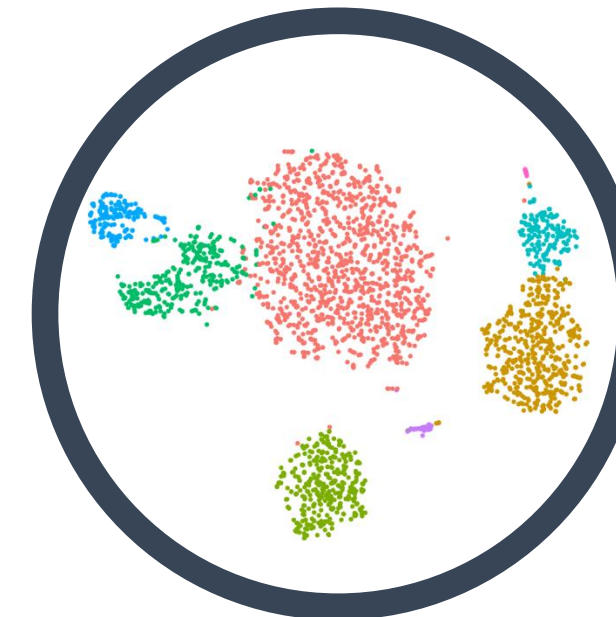
BIOINFORMATICS IN R

HIGH THROUGHPUT DATA



BIOINFORMATIC ANALYSIS

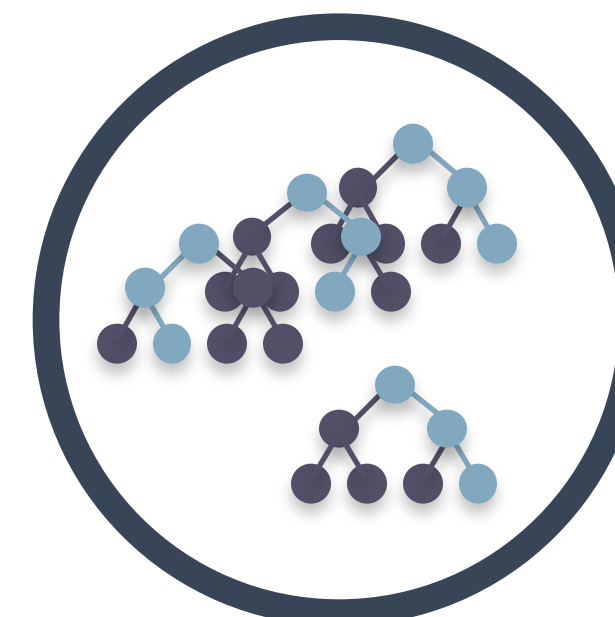
DIMENSIONALITY REDUCTION



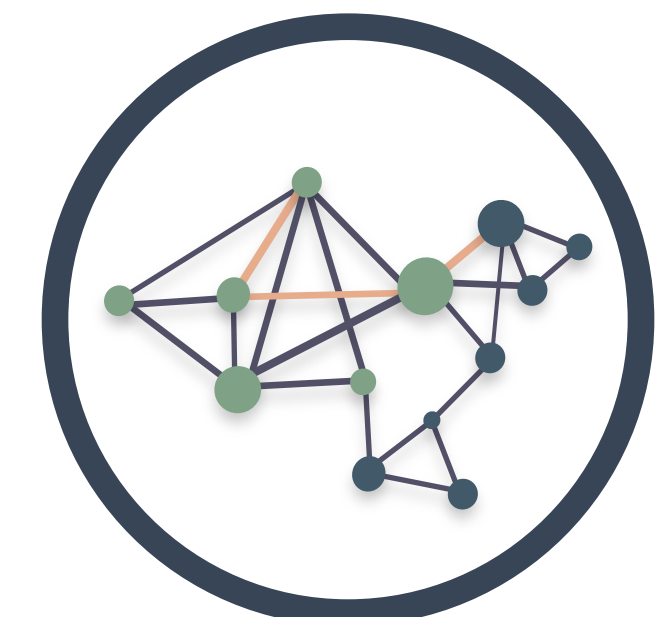
CLUSTERING

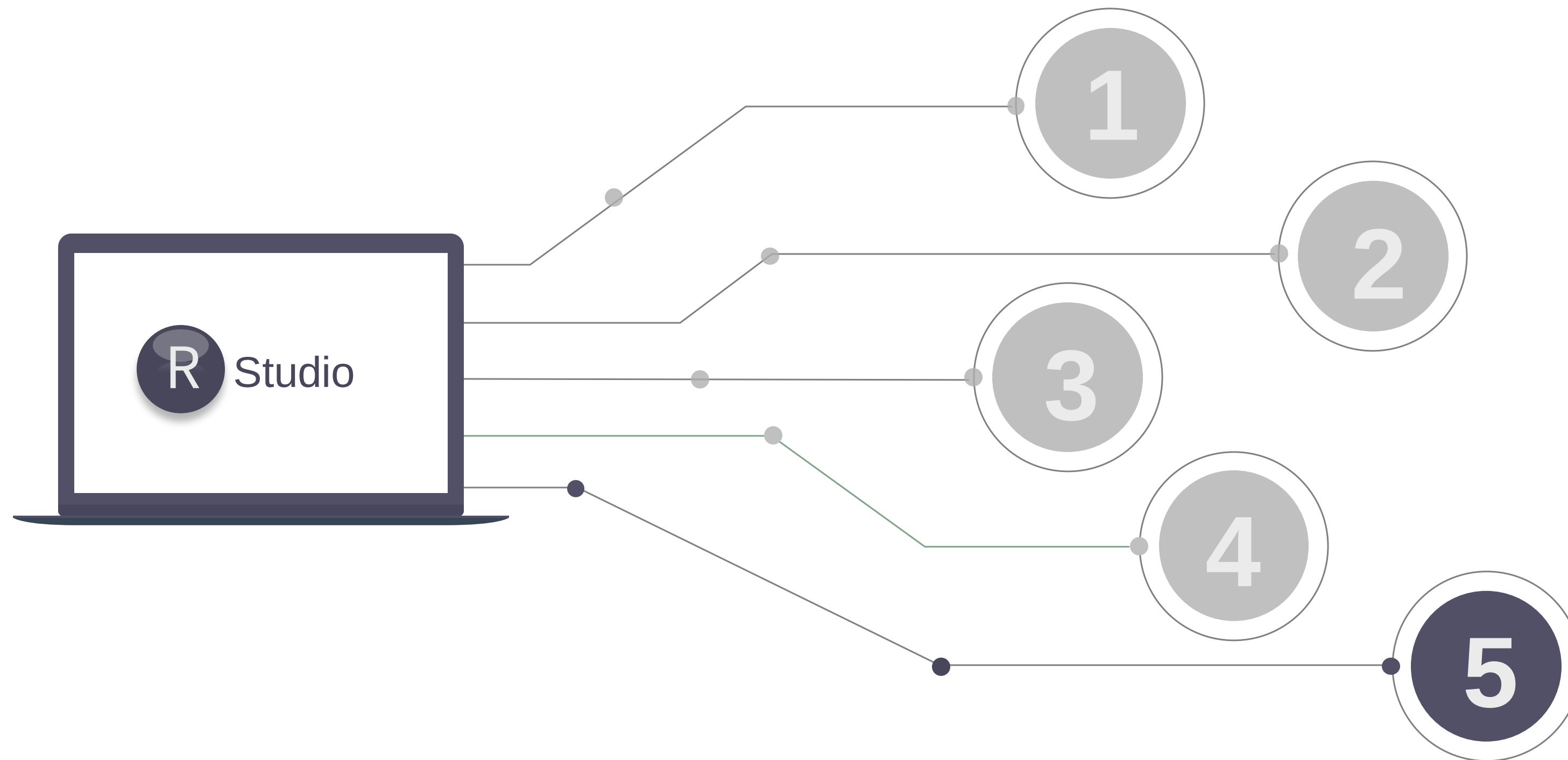


MACHINE LEARNING



NETWORK ANALYSIS





- Tidyverse
- ggplot
- Stats
- Quarto

Statistics in R

EXERCISE 5

THE TOP OF THE R ICEBERG



STATISTICAL ANALYSIS

Statistical models (linear, generalized, mixed, ...)

Statistical tests (t-test, chisq, anova, ...)

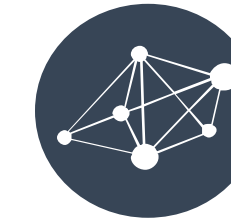
Survival analysis (Cox, Kaplan meier)



DATA MANGEMENT



EASY PLOTTING



BIOINFORMATIC ANALYSIS



WANT MO-R-E?

The Section for Biostatistics offers a number statistics-oriented R courses:

Spring

[Basic Statistics for Health Researchers \(Danish course\)](#)

ECTS: 9,0

[Epidemiological methods in medical research](#)

ECTS: 7

[Advanced topics in health research B](#)

ECTS: 2,8

[Statistical methods in bioinformatics](#)

ECTS: 3,5

[Statistical analysis of survival data](#)

ECTS: 4,9

<https://publichealth.ku.dk/about-the-department/biostat/>

[Programming and statistical modelling in R](#)

ECTS: 1,6

[Bayesian methods in biomedical research](#)

ECTS: 2,4

[Psychometric validation of patient reported outcome measures](#)

ECTS: 2,5

[Introduction to validation of patient reported outcome measures](#)

[Causal inference I](#)

ECTS: 2,5

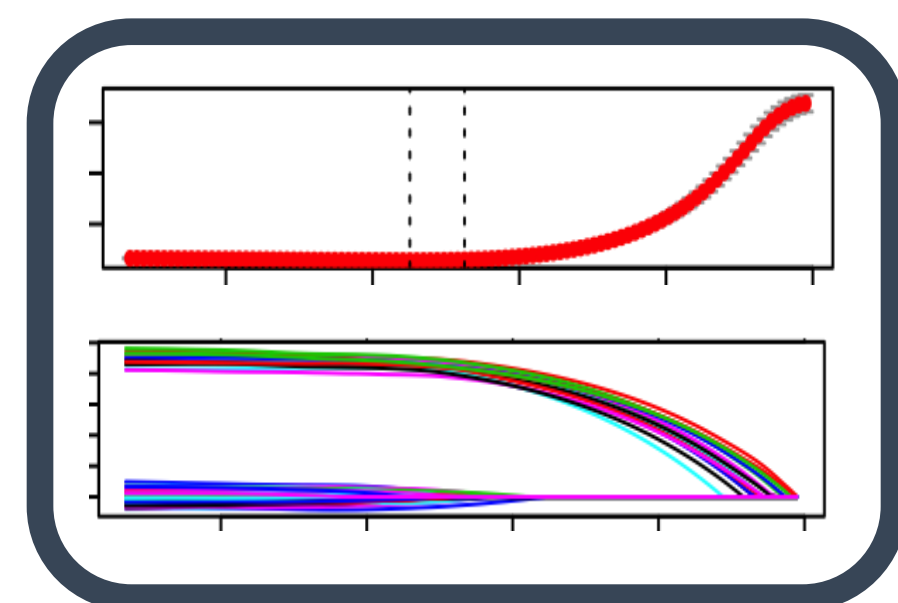
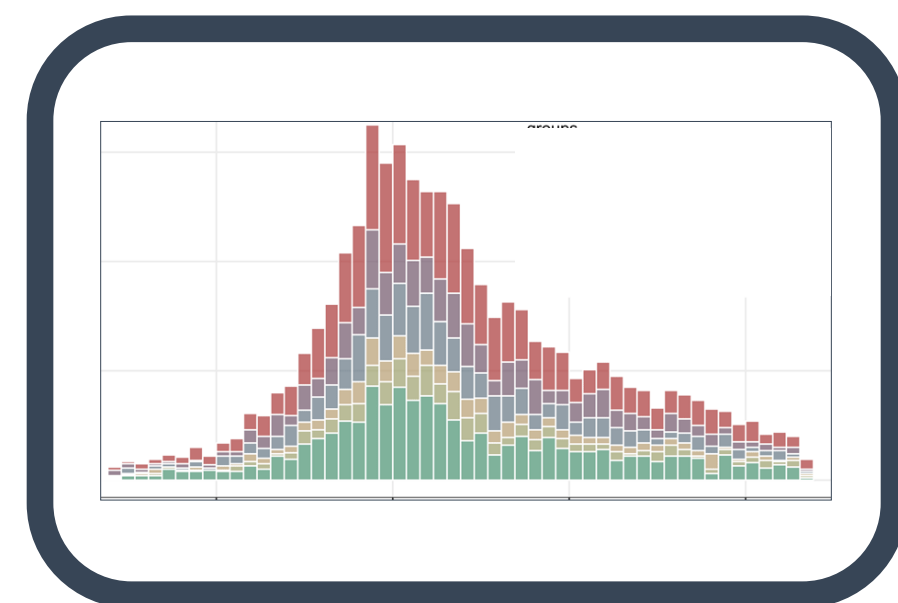
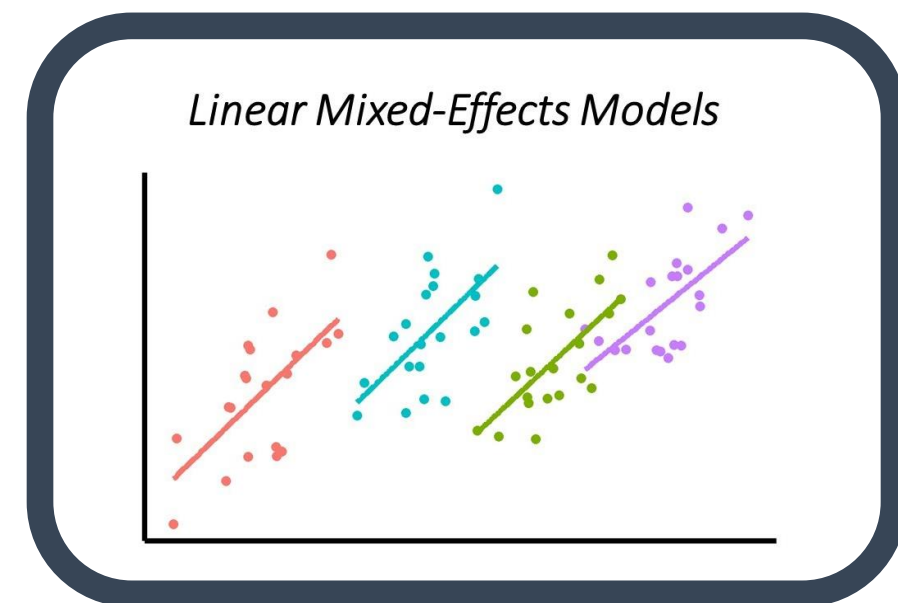
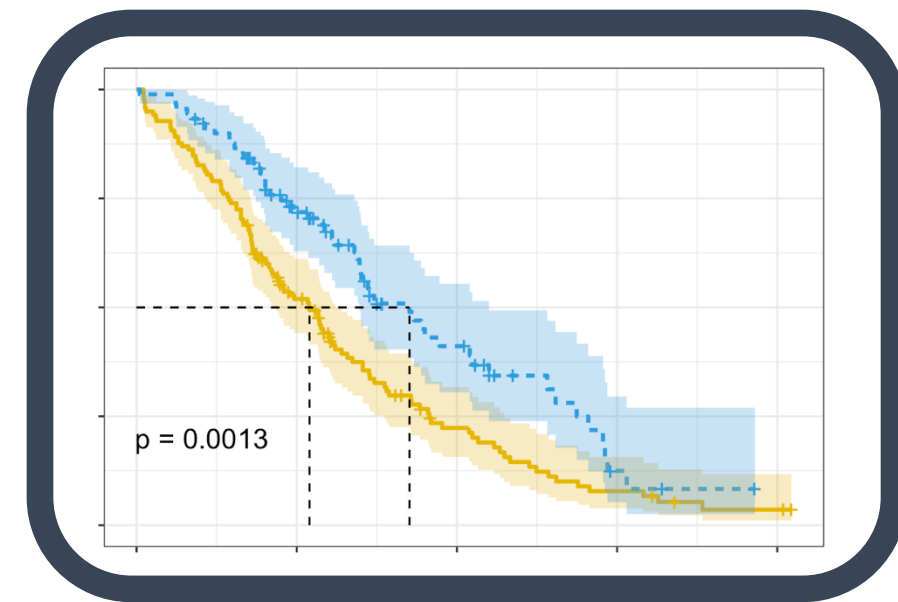
[Use of the statistical software R](#)

ECTS: 2,1

* These are screenshots. Go to the website and scroll down to 'Teaching'

TEASER

STATISTICS in R



Survival Analysis

survival: <https://rviews.rstudio.com/2017/09/25/survival-analysis-with-r/>

survminer: <https://cran.r-project.org/web/packages/survminer/survminer.pdf>
(<https://rpkgs.datanovia.com/survminer/>)

Mixed-Effects Models

lme4: <https://cran.r-project.org/web/packages/lme4/vignettes/lmer.pdf>

<https://cran.microsoft.com/snapshot/2017-08-01/web/packages/sjPlot/vignettes/sjplmer.html>

glmmTMB: <https://cran.r-project.org/web/packages/glmmTMB/index.html>

Epidemiological Analysis

Epi: <https://cran.r-project.org/web/packages/Epi/index.html>

pubh: <https://rviews.rstudio.com/2020/03/05/covid-19-epidemiology-with-r/>

https://cran.r-project.org/web/packages/incidence/vignettes/customize_plot.html

<https://rviews.rstudio.com/2020/03/05/covid-19-epidemiology-with-r/>

Elastic-Net Regression

(R

glmnet: <https://cran.r-project.org/web/packages/glmnet/glmnet.pdf>

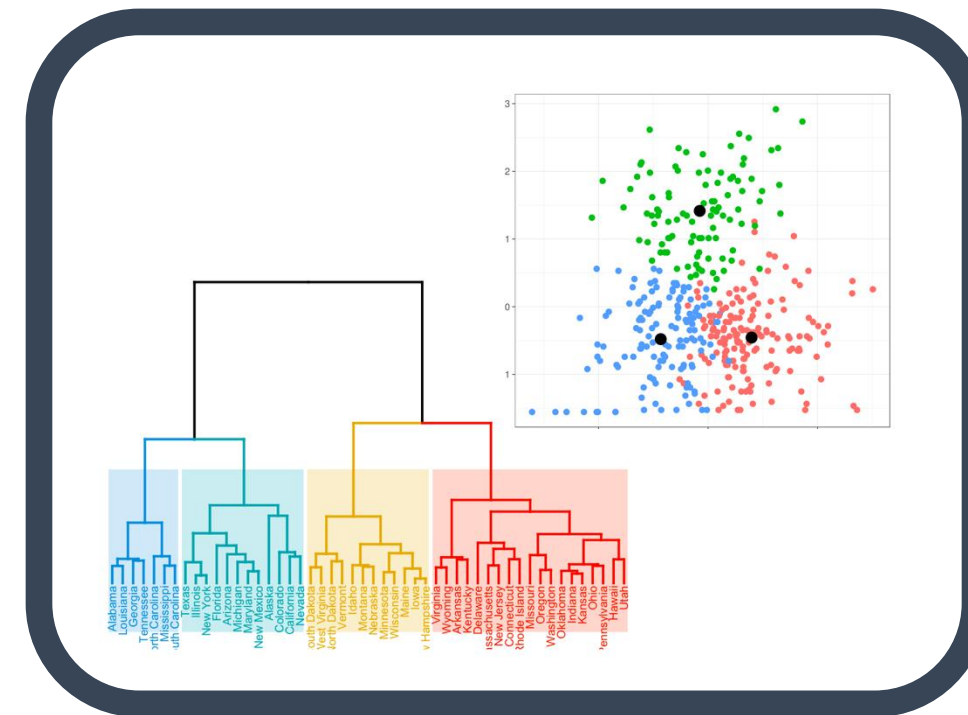
elasticnet: <https://cran.r-project.org/web/packages/elasticnet/elasticnet.pdf>

<https://www.datacamp.com/community/tutorials/tutorial-ridge-lasso-elastic-net>

TEASER Machine Learning

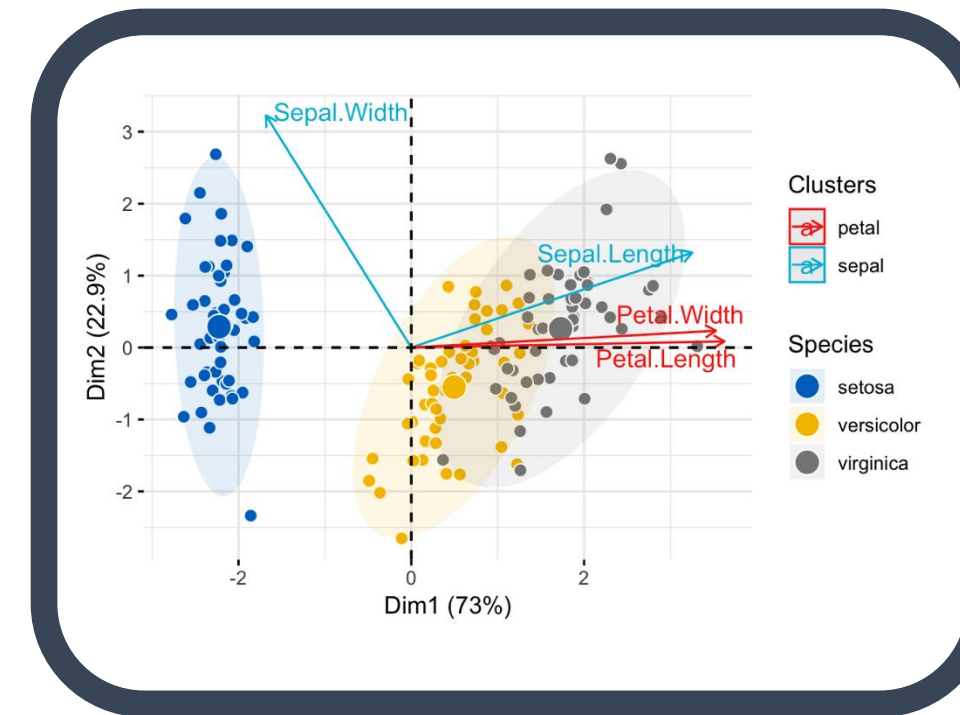
<https://lgatto.github.io/IntroMachineLearningWithR/an-introduction-to-machine-learning-with-r.html>

Clustering



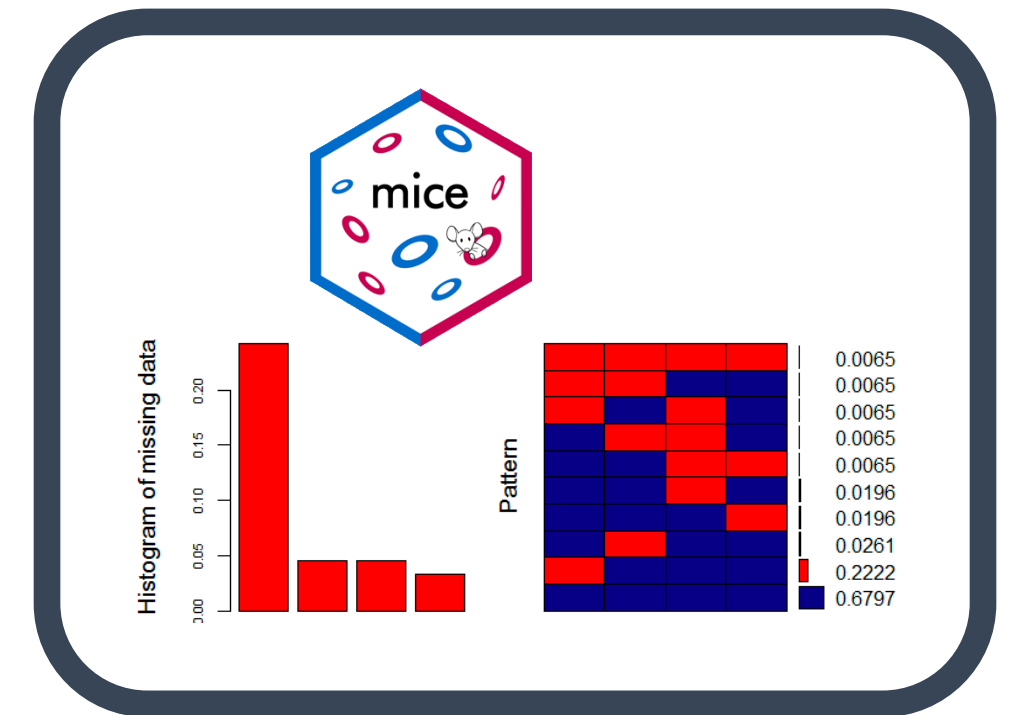
<https://statsandr.com/blog/clustering-analysis-k-means-and-hierarchical-clustering-by-hand-and-in-r/>

Feature Selection: PCA



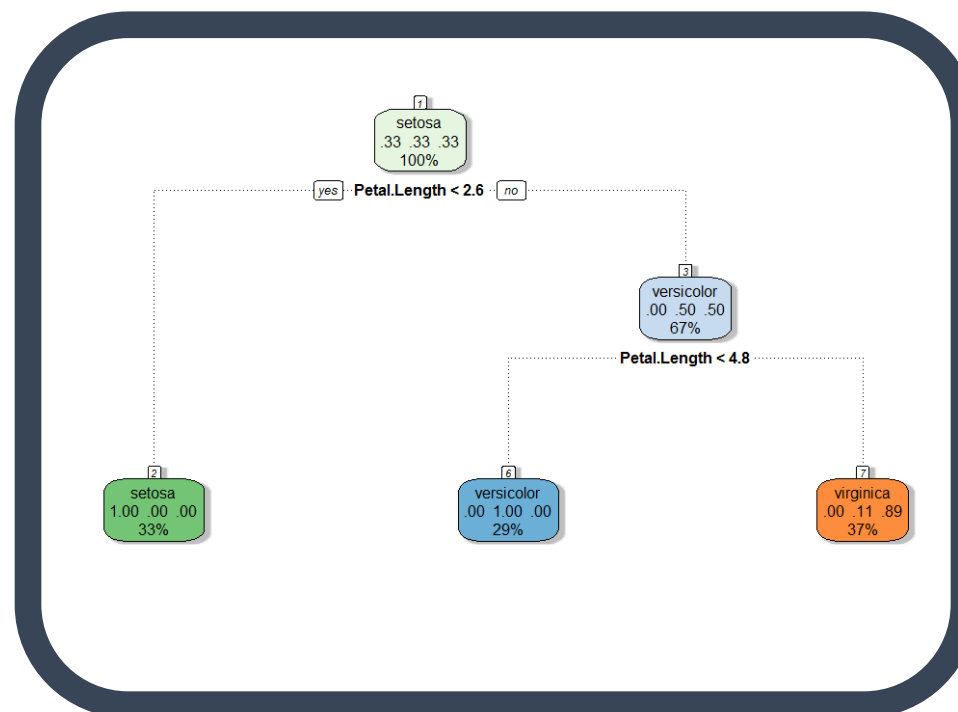
<https://bioconductor.org/packages/release/bioc/vignettes/PCAtools/inst/doc/PCAtools.html>

Missing Data



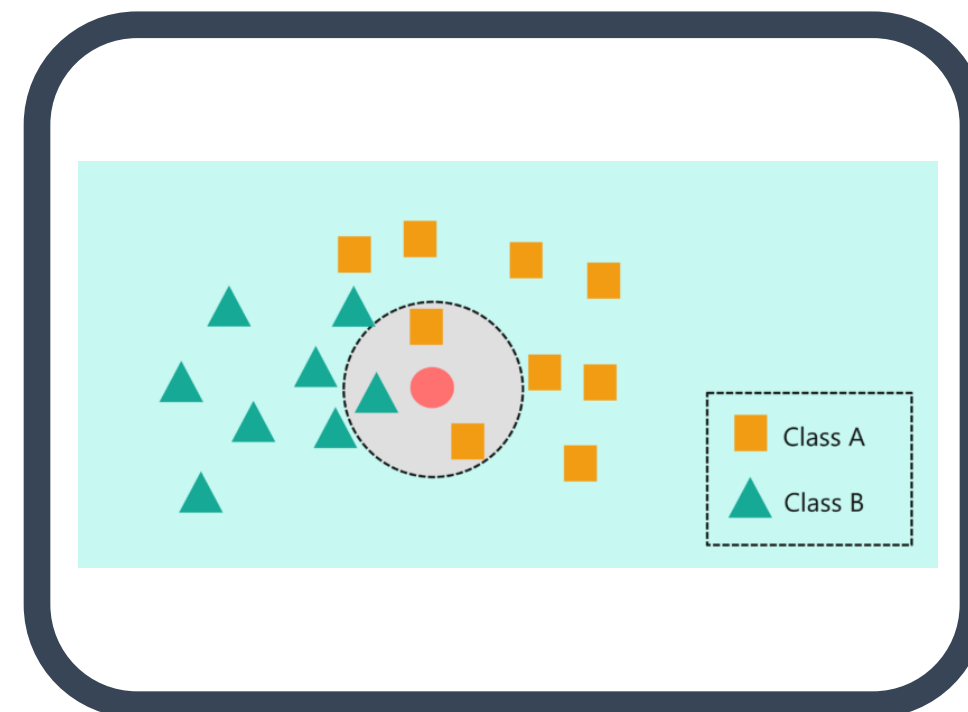
<https://amices.org/mice/>
<https://datascienceplus.com/imputing-missing-data-with-r-mice-package/>

Random Forest



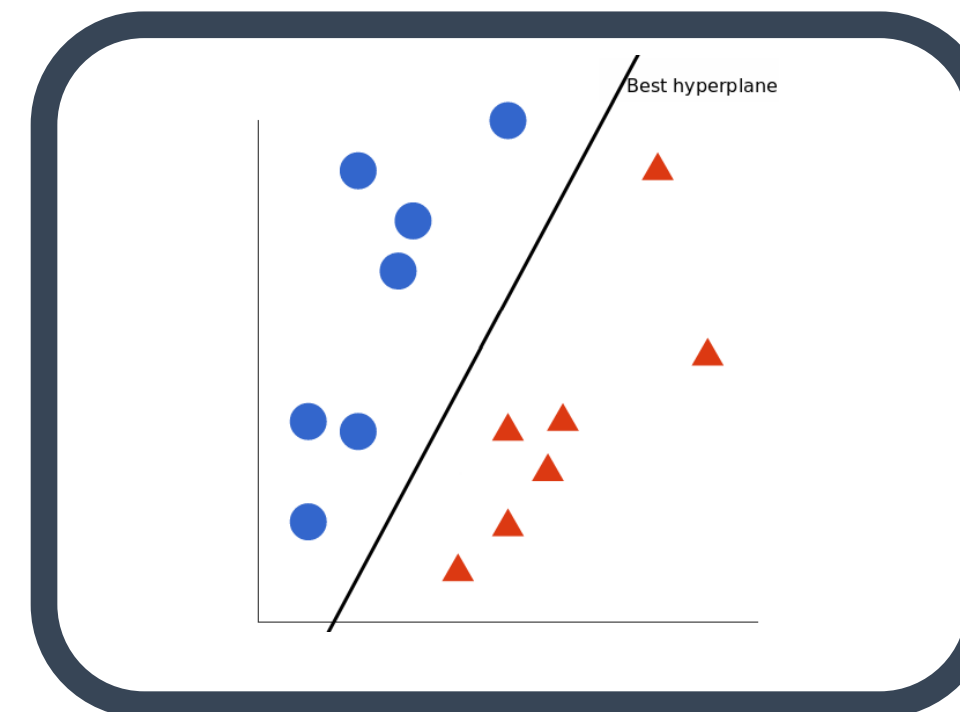
<https://www.blopig.com/blog/2017/04/a-very-basic-introduction-to-random-forests-using-r/>

kNN



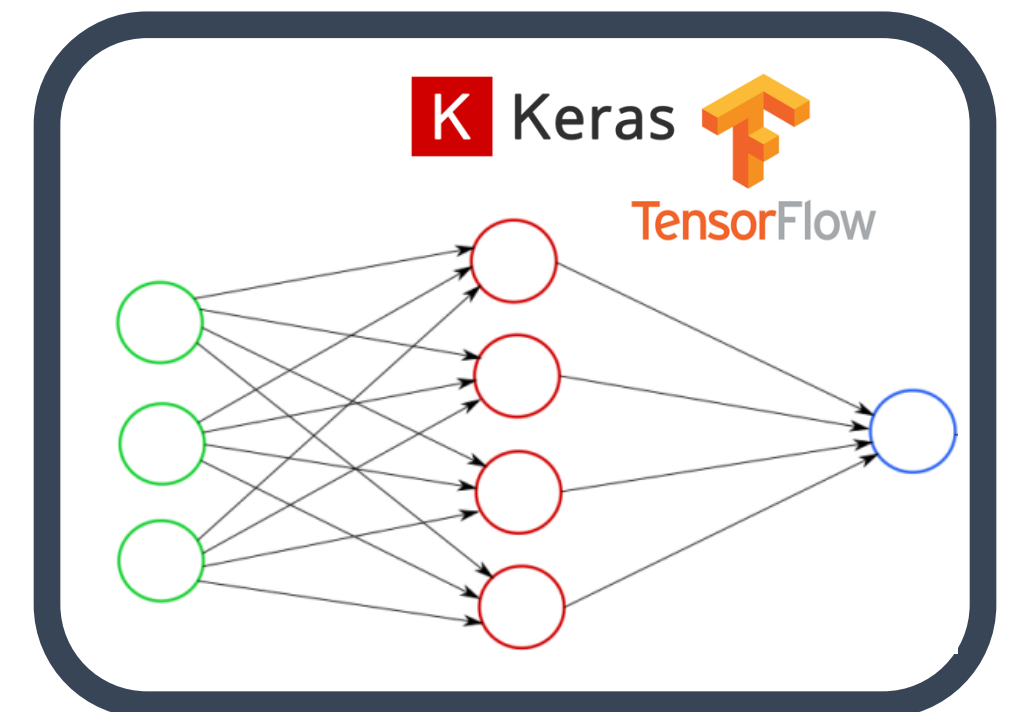
<https://www.edureka.co/blog/knn-algorithm-in-r/>

SVM



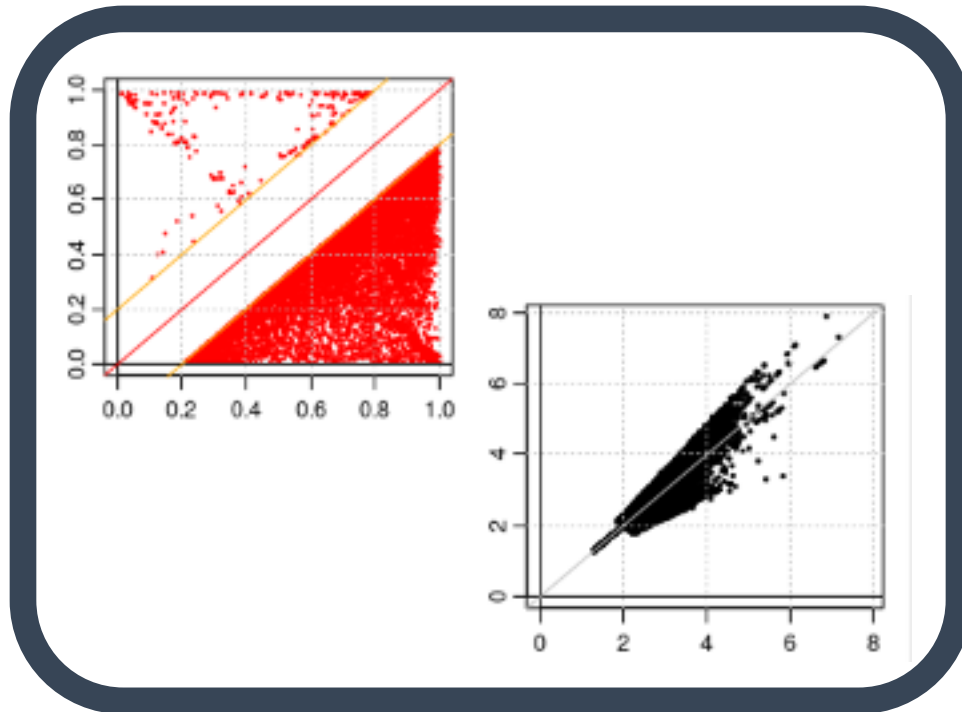
<https://cran.r-project.org/web/packages/e1071/vignettes/svm.doc.pdf>

Neural Networks



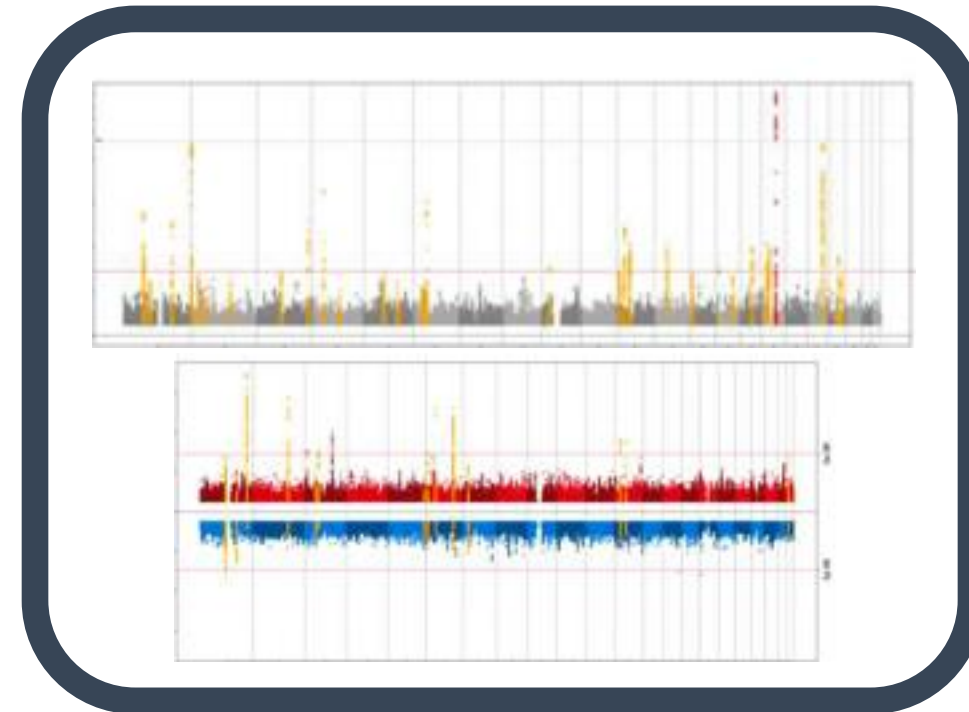
<https://keras.rstudio.com/>
<https://tensorflow.rstudio.com/>

GWAS - QC & Data Harmonization



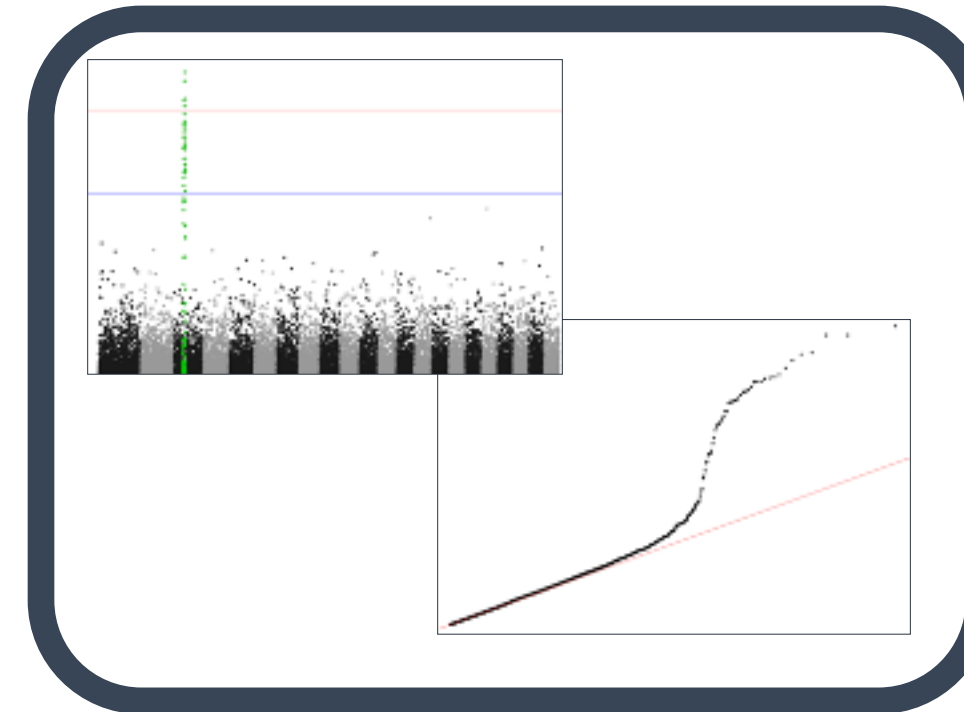
EasyQC: <https://www.uni-regensburg.de/medizin/epidemiologie-praeventivmedizin/genetische-epidemiologie/software/>

GWAS Data Management & Plots



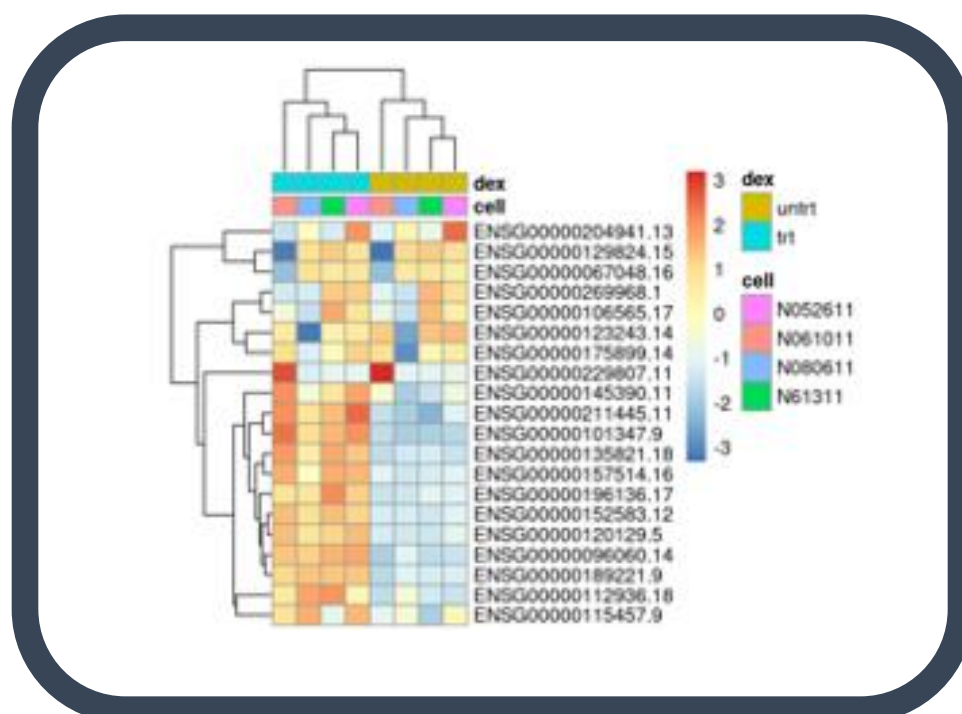
EasyStrata: <https://www.uni-regensburg.de/medizin/epidemiologie-praeventivmedizin/genetische-epidemiologie/software/>

More Plotting...



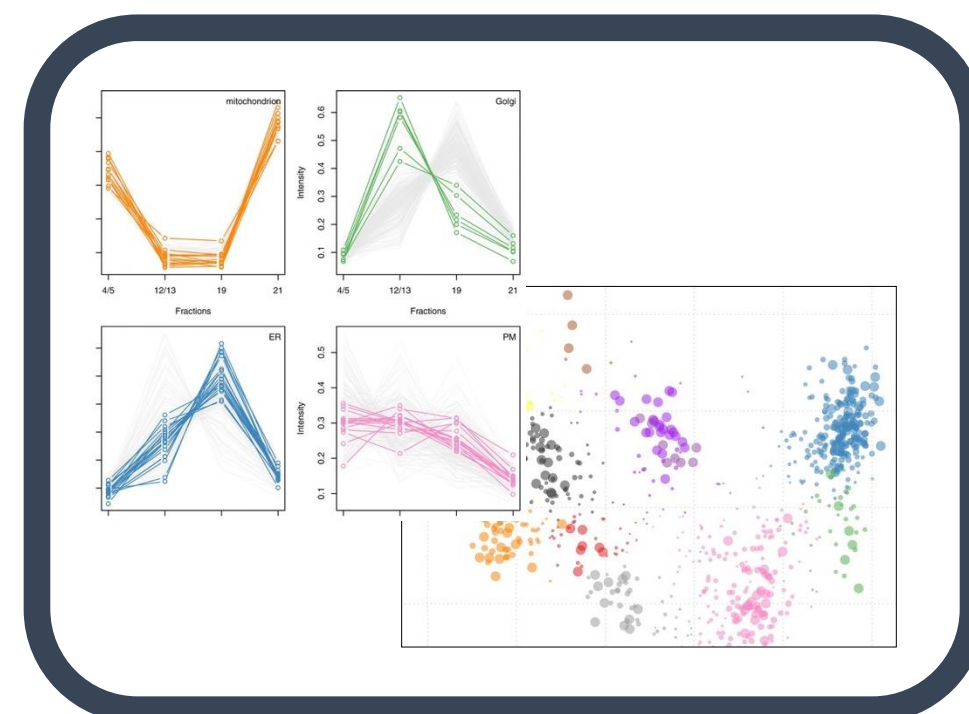
Manhattan and QQ plots:
<https://cran.r-project.org/web/packages/ggman/vignettes/ggman.htm>

Gene Expression Analysis



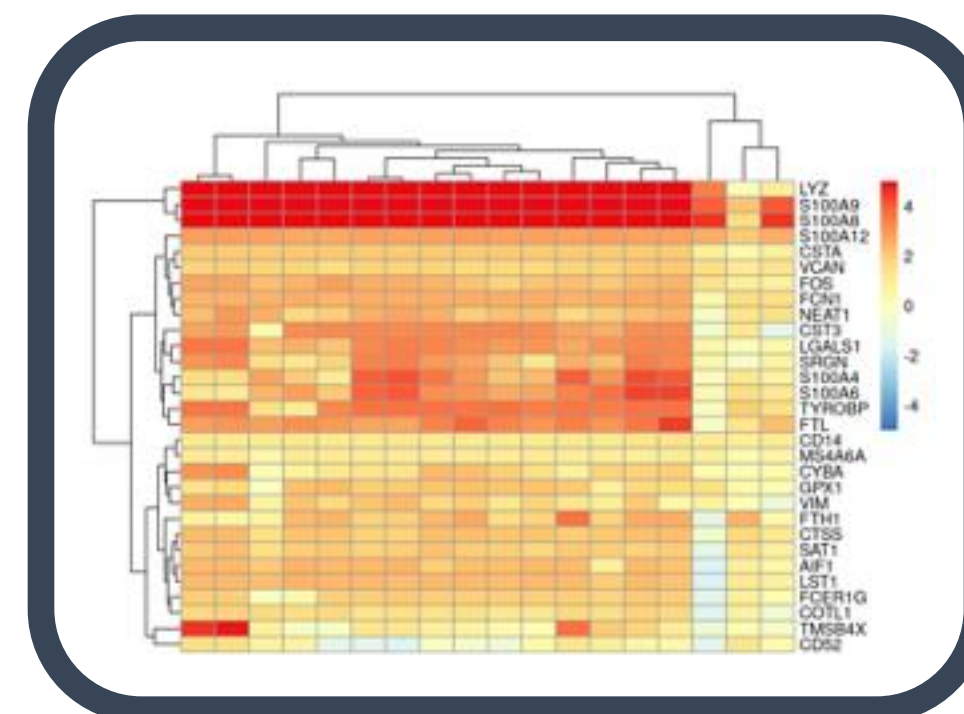
DESeq2, limma, EdgeR, etc.:
http://www.bioconductor.org/packages/release/BiocViews.html#_RNASeq

Proteomics Analysis



RforProteomics:
https://www.bioconductor.org/packages/release/BiocViews.html#_Proteomics

Single-Cell RNASeq



<https://cran.r-project.org/web/packages/e1071/vignettes/svmdoc.pdf>

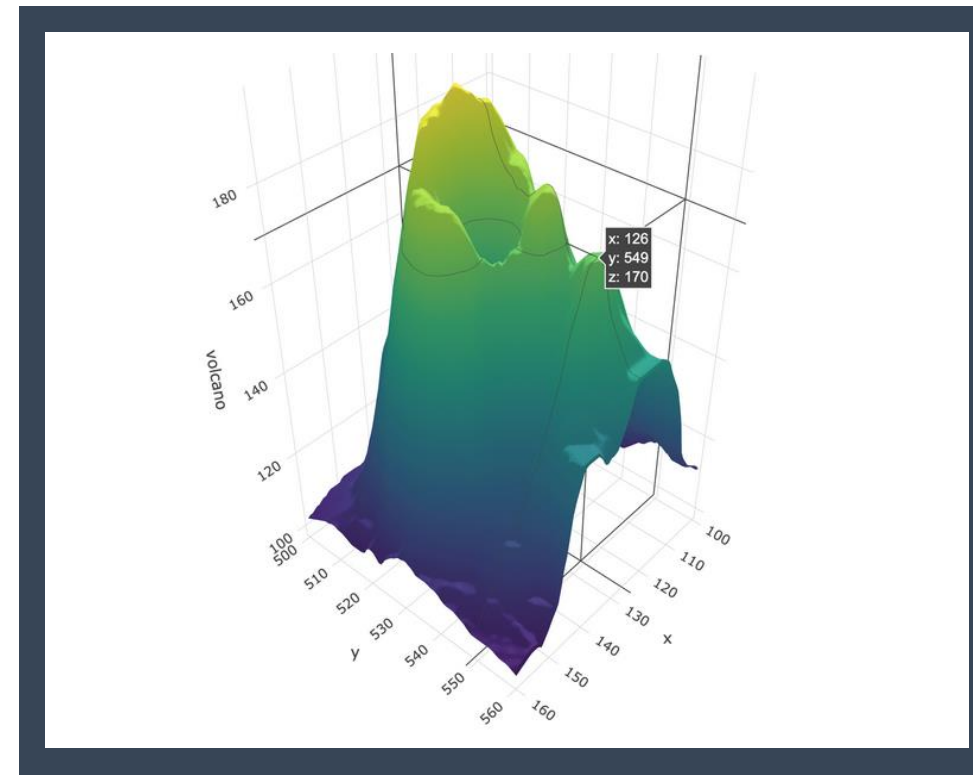
TEASER Omics Data

<http://www.bioconductor.org/packages/release/BiocViews.html>

FROM EXCEL TO R

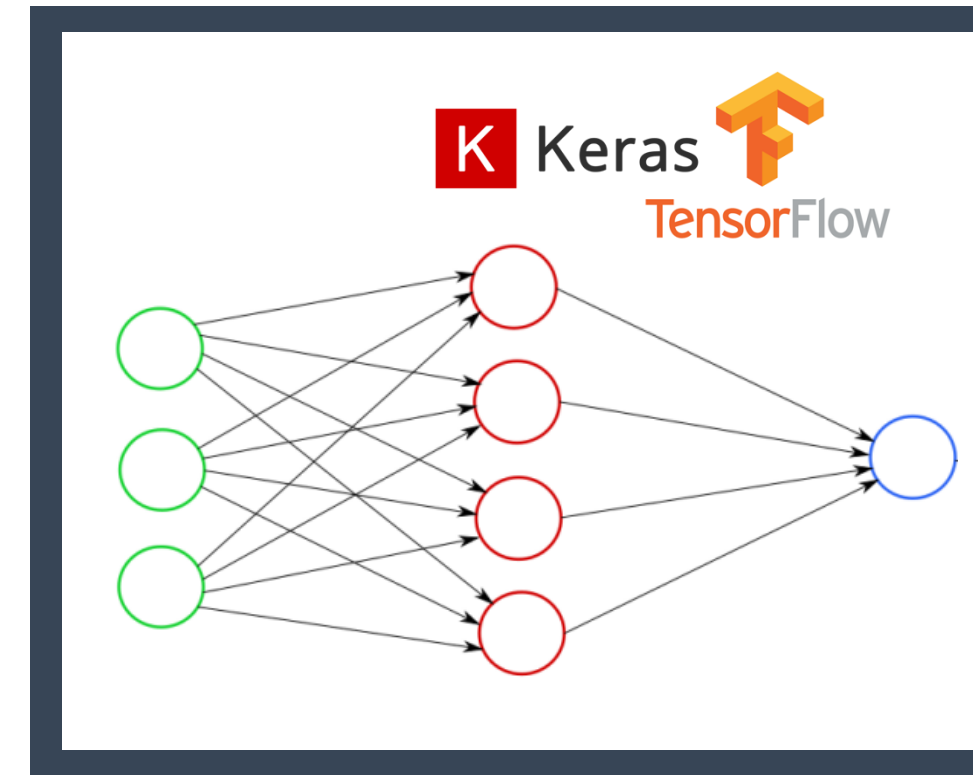
COOL STUFF IN R

PLOTTING IN 3D



<https://plotly-r.com/d-charts.html>

DEEP LEARNING



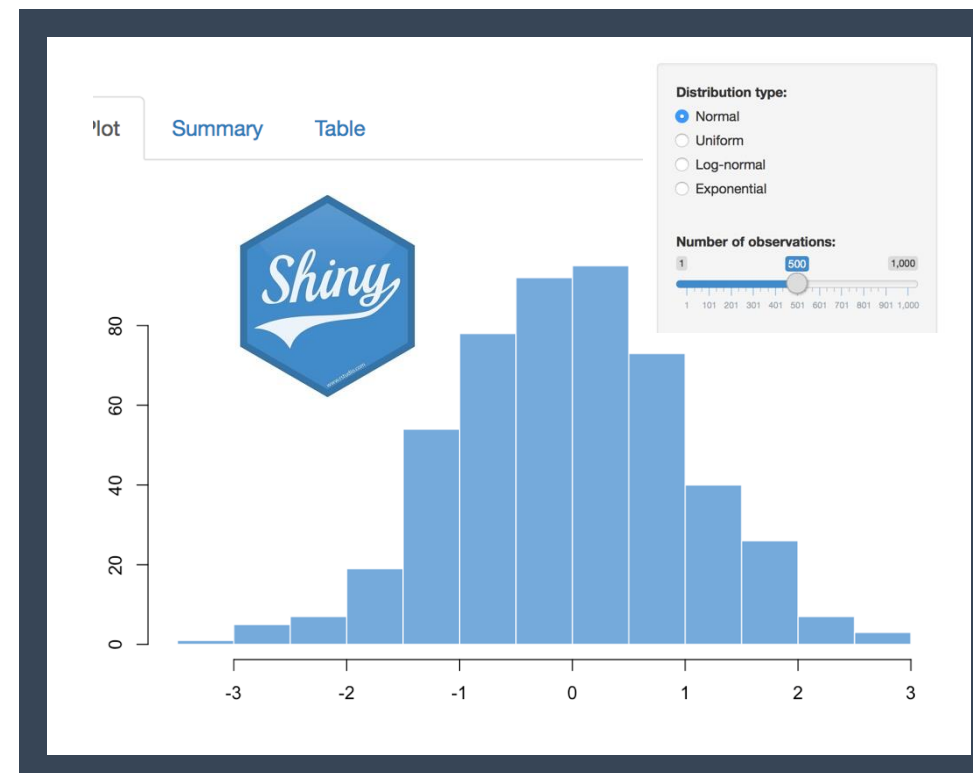
<https://keras.rstudio.com/>
<https://tensorflow.rstudio.com/>

BAYESIAN STATISTICS



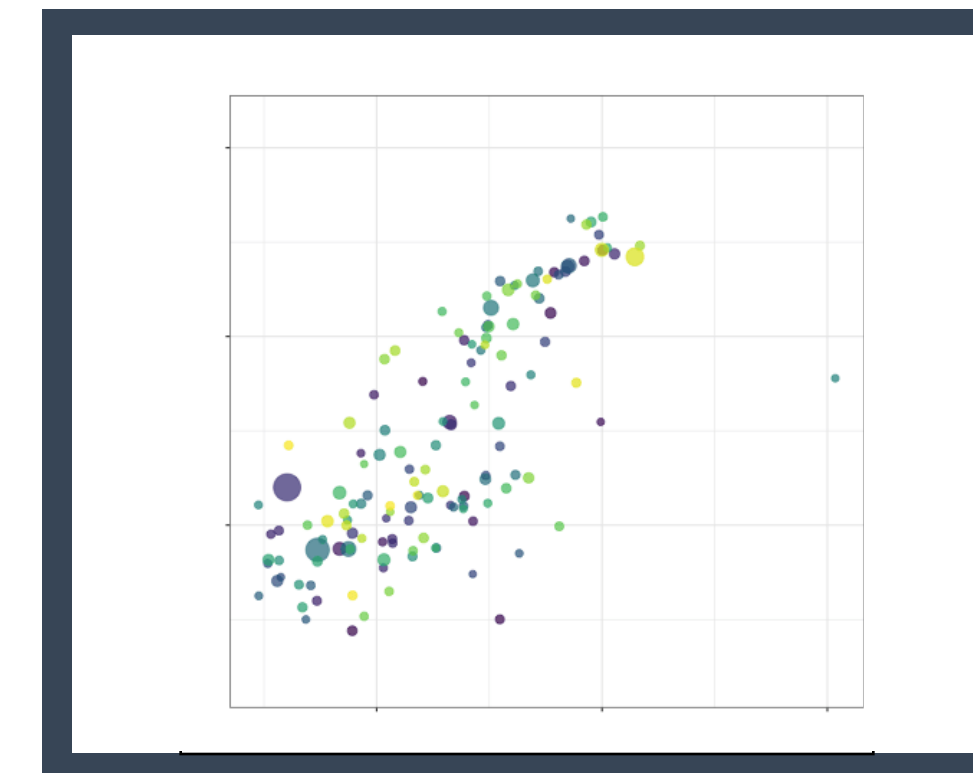
<https://mc-stan.org/users/interfaces/rstan>

WEBPAGE WITH R SHINY



<https://shiny.rstudio.com/>

INTERACTIVE PLOTS



<https://gganimate.com/articles/gganimate.html>

MAIL AND MESSAGES



<https://github.com/briandconnelly/pushoverr>

DATA SCIENCE WITH HEADS DATA LAB

- Drop-in Thursday (13-15)
- Consultations
- Commission Research & Supervision



Read more on HeaDS Website!

<https://heads.ku.dk>

- Events and Conferences
- Course Reminders



Flow HeaDS on LinkedIn!

<https://www.linkedin.com/company/ucph-heads/>

Health Data
Science
Seminar
& Social
Networking



TALKING HeaDS

Understanding the Effects of Missense Variants

Using Analyses of Protein Stability
and Conservation

with **Kresten Lindorff-Larsen**

27 May at 15:00
Panum, Faculty Club, 16.6.16

REGISTER



Supported by:



THANK YOU FOR LISTENING

